Learner's Workbook

Worksheets and exercises for the 40-hour CDF intensive

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
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DRAFT

Learner's Workbook

Worksheet 1.1: Introduction to Space Mission Design

Name:	____
Date:	____
Team:	____
Session:	1.1 -- Introduction to Space Mission Design

Part A: The 17 SE Processes -- SpaceCDF Mapping

For each NASA SE process (NPR 7123.1D), identify which SpaceCDF feature implements it. Write the tab name, button, or panel. If the process requires human judgment that the tool cannot automate, mark "Manual" in the last column.

Hint: Navigate through SpaceCDF's workflow steps (Need -> Concept -> Requirements -> Design) and explore each tab. Many processes map to more than one feature.

#	SE Process	SpaceCDF Feature(s)	Manual
1	Stakeholder Expectations Definition	____	______
2	Technical Requirements Definition	____	______
3	Logical Decomposition	____	______
4	Design Solution Definition	____	______
5	Product Implementation	____	______
6	Product Integration	____	______
7	Product Verification	____	______
8	Product Validation	____	______
9	Product Transition	____	______
10	Technical Planning	____	______
11	Requirements Management	____	______
12	Interface Management	____	______
13	Technical Risk Management	____	______
14	Configuration Management	____	______
15	Technical Data Management	____	______
16	Technical Assessment	____	______
17	Decision Analysis	____	______

Discussion question: Which processes are well-supported by the tool? Which ones fundamentally require human judgment and cannot be fully automated? Write your answer below:

Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase (Pre-A through F) it belongs to and which review gate marks the end of that phase:

Activity	NASA	Review Gate
"We need to monitor sea ice thickness in the Canadian Arctic"	______	______
Writing system-level "shall" requirements with verification methods	______	______
Selecting a specific reaction wheel vendor after evaluating 3 options	______	______
Running thermal-vacuum qualification tests on the flight model	______	______
Operating the satellite and producing daily data products	______	______
Decommissioning the satellite and performing passivation	______	______
Preliminary orbit trade study comparing SSO, ISS-orbit, and polar	______	______
Detailed finite element analysis of the primary structure	______	______

Hint: Refer to the lifecycle phase table in Session 1.1, Section 4.3. Ask yourself: "What level of design detail does this activity require?"

Part C: CDF Position Interaction Analysis

For each design conflict below, identify which engineering positions must interact to resolve it. Then briefly describe how you would approach the resolution.

Conflict 1: The payload requires 50 W but the solar array can only provide 30 W.

Positions involved: ________________________________

Resolution approach: ________________________________

Conflict 2: The reaction wheel vibration exceeds the payload's pointing stability requirement.

Positions involved: ________________________________

Resolution approach: ________________________________

Conflict 3: The selected orbit (700 km) requires a propulsive deorbit for debris compliance, but there is no mass budget remaining for a propulsion system.

Positions involved: ________________________________

Resolution approach: ________________________________

Conflict 4: The X-band downlink antenna pattern interferes with the payload's sensitive optical receiver.

Positions involved: ______________________________

Resolution approach: ______________________________

Part D: SpaceCDF Tool Exploration

Navigate through the SpaceCDF tool and answer each question:

1. How many KPI cards are shown on the Design Dashboard? List them by name:

2. In the Gate Review tab, what are the MCR exit criteria? Which are auto-evaluated?

3. What engineering positions are available when starting a collaborative session?

4. What workflow steps does SpaceCDF guide you through (from left to right)?

5. In the Positions tab, find your assigned position. What key questions does it list?

Part E: V-Model Sketch

In the space below, sketch the System-V model from memory. Label each level on both the left side (decomposition) and right side (integration). Draw the horizontal traceability arrows and label them.

Use this space for your sketch:

Notes & Reflections

Record key points from the group discussion, questions you want to follow up on, and connections to your mission concept:

Worksheet 1.2: Canadian Space Ecosystem & Regulatory Environment

Name:	______________________________
Date:	______________________________
Team:	______________________________
Session:	1.2 -- Canadian Space Ecosystem & Regulatory Environment

Part A: Canadian Space Industry Mapping

For your team's mission concept, identify potential Canadian industry partners at each tier. If no Canadian supplier exists for a capability, note what country you would source from and why.

Tier	Capability Needed	Canadian Candidate(s)	International Alternative	Justification
Tier 1 (Prime/Integration)	________________	________________	___________________	________________
Tier 2 (Subsystem)	________________	________________	___________________	________________
Tier 2 (Subsystem)	________________	________________	___________________	________________
Tier 3 (Component/Service)	________________	________________	___________________	________________
Tier 4 (R&D/Academic)	________________	________________	___________________	________________
	___	___		___

Question: What CSA funding program(s) might support your mission? Why?

Part B: RSSSA Compliance Assessment

Answer the following questions for your team's mission concept:

1. Does your mission involve any remote sensing of Earth (imaging, SAR, radiometry, etc.)?

2. If yes, will the satellite be operated from Canada?

3. What RSSSA conditions might apply? (Check all that apply)

Condition	Applies?	Notes
Priority access for Government of Canada	_________	___________________
Shutter control provisions	_________	___________________
Data disposition / handling plan	_________	___________________
System disposal plan	_________	___________________
Raw data protection	_________	___________________
Interoperability arrangements (foreign access)	_________	___________________

4. If your mission does NOT involve remote sensing, what other regulatory requirements apply? (e.g., spectrum licensing, export control)

Part C: Spectrum Licensing Plan

Identify the frequency bands your mission will use and the licensing path for each:

Link	Band	Frequency (GHz)	Data Rate	Use (TT&C / Payload / ISL)	Licensing Path
Uplink	__________	___________	__________	___________	___________
Downlink (TT&C)	__________	___________	__________	___________	___________
Downlink (Payload)	__________	___________	__________	___________	___________
Other	__________	___________	__________	___________	___________

Hint: For a typical CubeSat, TT&C uses S-band (2.0--2.3 GHz) and payload data uses X-band (8.025--8.4 GHz). The licensing path for each is: ISED application -> ISED technical review -> ITU filing (API) -> ITU coordination -> ISED licence issuance.

Question: How early in your project should you begin the spectrum licensing process? Why?

Part D: Export Control Classification

List the 3 most critical components in your preliminary mission concept. For each, identify the likely export control classification:

Component	Country of Origin	Likely Classification	Licence Required?
_______________________	___________	ECCN: ___________ / USML Cat: ___________	___________
_______________________	___________	ECCN: ___________ / USML Cat: ___________	___________
_______________________	___________	ECCN: ___________ / USML Cat: ___________	___________

Hint: Most commercial-off-the-shelf (COTS) space components from US manufacturers are classified under EAR (ECCN 9A515 for spacecraft parts). Military-grade or specifically designed items may fall under ITAR (USML Category XV). Canadian-origin components are subject to EIPA if they appear on the Export Control List.

Question: If your mission includes a US-origin star tracker and you plan to launch on an Indian PSLV, what export control issues arise?

Part E: Regulatory Timeline

Based on your mission concept, estimate when each regulatory activity should begin and how long it will take. Mark the activities on the timeline below:

Regulatory Activity	Start Phase	Duration	Critical Path?
RSSSA licence application	__________	__________	__________
ISED spectrum application	__________	__________	__________
ITU coordination	__________	__________	__________
Export permit(s)	__________	__________	__________
Launch provider interface agreement	__________	__________	__________

Question: Which regulatory activity is most likely to be on the critical path for your mission? Why?

Notes & Reflections

Record key points from the discussion, regulatory questions you need to research further, and implications for your mission design:

Worksheet 1.3: International Standards for Space

Systems

Name:	____
Date:	____
Team:	____
Session:	1.3 -- International Standards for Space Systems

Part A: Standard Applicability Matrix

For your team's mission concept, determine which ECSS and NASA standards apply. For each standard, decide whether it is fully applicable, partially applicable (with tailoring), or not applicable. Justify your decision.

Standard	Branch	Applicability	Tailoring Notes / Justification
ECSS-M-ST-10C (Project Management)	M	__________	____________________________________
ECSS-M-ST-40C (Configuration Management)	M	__________	____________________________________
ECSS-M-ST-80C (Risk Management)	M	__________	____________________________________
ECSS-E-ST-10C (Systems Engineering)	E	__________	____________________________________
ECSS-E-ST-20C (Electrical & Electronic)	E	__________	____________________________________
ECSS-E-ST-32C (Structures)	E	__________	____________________________________
ECSS-E-ST-40C (Software)	E	__________	____________________________________
ECSS-E-ST-50C (Communications)	E	__________	____________________________________
ECSS-Q-ST-10C (Product Assurance)	Q	__________	____________________________________
ECSS-U-AS-10C (Debris Mitigation)	U	__________	____________________________________
ISO 24113 (Debris Mitigation)	--	__________	____________________________________
NPR 7123.1D (SE Processes)	--	__________	____________________________________

Hint: A university CubeSat mission would typically invoke ECSS-E-ST-10C with heavy tailoring, ECSS-U-AS-10C fully, and might waive ECSS-Q-ST-10C (product assurance) due to cost constraints. A CSA-funded mission would typically require more complete ECSS compliance.

Question: What is the minimum set of standards your mission MUST comply with regardless of mission class?

Part B: CDS Rev 14 Compliance Checklist

If your mission uses a CubeSat form factor, complete this compliance checklist. If not, skip to Part C.

CDS Requirement	Your Design Value	Compliant?	Notes
Unit dimensions (100 x 100 x 113.5 mm per U)	__________	__________	__________
Mass per U (<= 2.0 kg)	__________	__________	__________
Total mass (for your form factor)	__________	__________	__________
Rail material (hard-anodised Al)	__________	__________	__________
CG within 2 cm of geometric centre	__________	__________	__________
No protrusions beyond envelope during launch	__________	__________	__________
RF silence for 30 min after deployment	__________	__________	__________
Deployment switches on all deployables	__________	__________	__________
Propulsion inhibits (3 independent, if applicable)	__________	__________	__________

Hint: Use SpaceCDF's Compliance panel to check CDS compliance. The tool will flag non-compliant parameters automatically based on your mass and dimensions entries.

Part C: Debris Mitigation Compliance

Complete the following debris mitigation assessment for your mission orbit:

1. Planned orbit altitude: __________ km
2. Estimated natural orbital lifetime (without propulsion):

Use the rule of thumb: \tau_years ~= (h - 200)/(30) x (m/A)/(50) where h is altitude (km), m is mass (kg), A is cross-section area (m^2)

h = ______________________________

m/A = ______________________________

\tau = ______________________________

3. Compliance assessment:

Requirement	Limit	Your Mission	Compliant?
IADC post-mission lifetime	<= 25 years	__________	__________
FCC post-mission lifetime (2024+)	<= 5 years	__________	__________
ISO 24113 disposal reliability	>= 90%	__________	__________

4. If non-compliant with the 5-year FCC rule, what options exist?

____

____

____

5. Passivation plan (list all stored energy sources and how you will deplete them at end of life):

Energy Source	Passivation Method
Battery	______________________________

Pressure vessels (if any)	____
Reaction wheels	____
RF transmitter	____
Other: __________	____

Part D: Structural Design Margins

If your mission has preliminary structural design information, calculate the margin of safety:

Use the formula: MoS = (sigma_allowable)/(FoS x sigma_applied) - 1

Worked example: For Al 7075-T6 (sigma_yield = 503 MPa), applied stress = 200 MPa, FoS (yield) = 1.25:

MoS = (503)/(1.25 x 200) - 1 = (503)/(250) - 1 = 1.012

This is a positive margin (compliant).

Your calculation:

Material: ____

sigma_applied = ____

MoS = ____

Part E: SpaceCDF Compliance Panel

Navigate to the Compliance panel in SpaceCDF and answer:

1. What is the debris mitigation compliance status for your mission?

---

2. What orbital altitude would be needed to comply with the FCC 5-year rule WITHOUT propulsion? (Use the tool's orbital lifetime calculator)

---

3. What standards has the tool automatically suggested for your mission type?

---

4. Are there any compliance flags (warnings or errors) on your current design?

Notes & Reflections

Record key points about standards applicability, compliance challenges, and design implications for your mission:

Worksheet 1.4: Mission Needs, Trade Studies & Concept of Operations

Name:	________________________________
Date:	________________________________
Team:	________________________________
Session:	1.4 -- Mission Needs, Trade Studies & ConOps

Part A: Stakeholder Analysis Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, priority, and influence level.

#	Stakeholder	Role	Key Needs	Constraints Imposed	Priority (P/S/T)	Influence (H/M/L)
1	___________	___________	___________________________	___________________________	_______	_______
2	___________	___________	___________________________	___________________________	_______	_______
3	___________	___________	___________________________	___________________________	_______	_______
4	___________	___________	___________________________	___________________________	_______	_______
5	___________	___________	___________________________	___________________________	_______	_______
6	___________	___________	___________________________	___________________________	_______	_______

P = Primary (must satisfy), S = Secondary (should satisfy), T = Tertiary (nice to have)

H = High influence, M = Medium, L = Low

Question: Which stakeholder has the most influence over your mission? Do any stakeholders have conflicting needs? If so, how might you resolve the conflict?

Part B: Problem Statement

Write your team's mission problem statement. It must answer all four questions: What is the problem? Who is affected? What is the impact? What constraints exist?

Problem Statement:

Peer Review Checklist (exchange with another team):

Criterion	Pass?	Feedback
Describes WHAT is needed, not HOW to achieve it?	_______	____________________________________
Identifies the capability gap?	_______	____________________________________

Names specific affected stakeholders?	______	____________________________
Quantifies the impact of not solving it?	______	____________________________
States constraints (budget, schedule, regulatory)?	______	____________________________
Free of solution language? (No specific satellite type, orbit, component named)	______	____________________________

_____

Part C: Mission Objectives

Write 2--3 mission objectives. Score each against the quality checklist.

Objective 1: ____________________

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	______	_________________________________
Has measurable criterion with number + unit?	______	_________________________________
Achievable with current technology (TRL >= 4)?	______	_________________________________
Relevant to the problem statement?	______	_________________________________
Traceable to a specific stakeholder need?	______	_________________________________
Classified as primary or secondary?	______	_________________________________

Objective 2: ____________________

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	______	_________________________________
Has measurable criterion with number + unit?	______	_________________________________
Achievable with current technology (TRL >= 4)?	______	_________________________________
Relevant to the problem statement?	______	_________________________________
Traceable to a specific stakeholder need?	______	_________________________________
Classified as primary or secondary?	______	_________________________________

Objective 3 (optional): ____________________

Part D: MoE / MoP / TPM Trace

For your primary objective, trace the measurement hierarchy:

Objective: ____________________

MoE (Measure of Effectiveness -- operational effectiveness from user perspective):

MoP (Measure of Performance -- technical performance of the system):

TPM (Technical Performance Measure -- design parameter tracked during development):

Hint: If your objective is "Provide 10 m GSD multispectral imagery...", then:

- MoE might be "% of target area with usable (<30% cloud) imagery within 5 days"
- MoP might be "GSD <= 10 m at nadir, SNR >= 100 in all bands"
- TPM might be "current estimate of aperture diameter (target: 15 cm), imager mass (target: 1.5 kg)"

Part E: Trade Study -- Space vs Non-Space

Complete a trade study evaluating whether a dedicated satellite is the right solution for your mission need.

Decision statement: ____________________

Alternatives:

#	Alternative	Description
1	______________________	____
2	______________________	____
3	______________________	____
4	Do nothing / status quo	____

Criteria and Weights (must sum to 1.0):

Use pairwise comparison or direct assignment. Show your weight derivation below.

Criterion	Weight	Direction (higher/lower is better)
______________________	_____	________
______________________	_____	________
______________________	_____	________
______________________	_____	________
______________________	_____	________
Total	1.00	

Weight derivation (pairwise comparison table or notes):

Scoring Matrix (normalised 0--1):

Alternative	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Weighted Total
1: ____________	______	______	______	______	______	______
2: ____________	______	______	______	______	______	______
3: ____________	______	______	______	______	______	______
4: Do nothing	______	______	______	______	______	______

Recall: Weighted Total = Sum w_c x s(a,c)

Result: The highest-scoring alternative is: ______________________________

Decision statement (1 paragraph -- include top 2 alternatives and why one was selected over the other):

____

____

____

____

____

Sensitivity check: If you increase the weight of your second-most-important criterion by 20% (and re-normalise), does the ranking change?

____

____

____

Part F: Concept of Operations Outline

Document your preliminary ConOps following the NASA SEH Appendix S structure:

1. Mission Architecture:

Space segment: ______________________________

Ground segment: ______________________________

User segment: ______________________________

2. Orbit-Average Power Budget:

Calculate using: P_avg = Sum P_mode,i x DC_i

Mode	Power (W)	Duty Cycle (fraction of orbit)	P x DC (W)

Idle / Housekeeping	___________	___________	___________
Science / Imaging	___________	___________	___________
Downlink	___________	___________	___________
Eclipse	___________	___________	___________
		Orbit-average power:	___________ W

Solar array requirement: P_SA = ______________________

Show your calculation:

____

____

3. Data Budget:

Daily data generation: ______________________

Downlink capacity per pass: ______________________

Passes per day: ______________________

Daily downlink capacity: ______________________

Balance (downlink >= generation)? ______________________

____

Part G: SpaceCDF Tool Entry

In SpaceCDF, enter your mission need (Step 1) and concept (Step 2). Then answer:

1. In Step 1, did the tool auto-detect your mission type from the objective text? What type did it suggest?

____

2. In Step 2, run the mission trade analysis. Does the tool's recommendation match your manual trade study result from Part E?

____

____

3. In the ConOps tab, build your architecture diagram. How many systems did you define?

____

4. In the Trade Studies tab, run an orbit selection trade with at least 3 orbit options. What orbit did it recommend?

____

Notes & Reflections

Record key decisions made today, open questions, and what you want to investigate further:

Worksheet 2.1: The System-V and Requirements Engineering

Name:	________________________________
Date:	________________________________
Team / Mission:	________________________________
Role / Position:	________________________________

SpaceCDF Tab: Requirements

Key Equations Reference

> *Traceability Coverage:*
>
> *$Coverage = \fracN_verified requirementsN_total requirements x 100%$*
>
> *Target: 100% -- every requirement needs a parent trace, a verification method, and a responsible subsystem.*

> *Requirement Quality (SMART):*
>
> *Specific -- Measurable -- Achievable -- Relevant -- Traceable*

> *Verification Methods (ATID):*
>
> *Analysis -- Test -- Inspection -- Demonstration*

Part A: System-V Mapping (10 min)

For your mission, identify one element at each level of the V-model:

V-Level	Left Branch (Decomposition)	Right Branch (Verification)
Stakeholder Need		
Mission Requirement		
System Requirement		
Subsystem Requirement		

Part B: Requirement Quality Assessment (15 min)

Evaluate each requirement using the SMART checklist and NASA Appendix C criteria. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	WHAT or	Rewrite (if needed)
1	"The spacecraft shall operate at 500 km altitude"							
2	"The system shall achieve GSD <= 10 m at nadir"							
3	"The EPS shall use triple-junction GaAs cells"							
4	"The system shall have a design lifetime of at least 3 years"							
5	"The structure shall be made of aluminium 7075"							

Part C: Requirement Hierarchy and Traceability (15 min)

For your mission, write one requirement at each level. Draw the traceability chain.

Stakeholder Objective (traces FROM): ______________________

**Mission Requirement (MR- ______________________

**System Requirement (SR- ______________________

**Subsystem Requirement (SSR- ______________________

Traceability diagram (draw arrows showing parent-child relationships):

Part D: Splitting Compound Requirements (10 min)

Split this compound requirement into individually testable statements:

> *"The system shall provide 10 m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."*

#	Split Requirement (one concern each)	Type (F/P/I/C/E)	Verification (A/T/I/D)	Phase
1				
2				
3				
4				
5				

Part E: SpaceCDF Requirements Generation (20 min)

1. In SpaceCDF, click "Generate from Objectives." How many requirements were generated? ____

2. Use the Level filter -- count at each level:
- Mission: ____   System: ____   Subsystem: ____

3. Classify 5 requirements by type:

4. Find one requirement that says HOW (implementation) not WHAT. Write it and your correction:

- Original: ____
- Corrected: ____

5. Accept at least 5 requirements and reject at least 1. Record your decisions:

6. For 5 key requirements, assign verification method and phase:

Notes & Reflections

Worksheet 2.2: Functional Decomposition and Allocation

Req ID	Requirement Text (abbreviated)	Type (Functional / Performance / Interface / Constraint / Environmental)	
Req ID	Action (Accept/Edit/Reject)	Rationale	
Req ID	Verification Method (A/T/I/D)	Phase (B/C/D/E)	Rationale for Method
Name:	______________________________ ____________		
Date:	______________________________ ____________		
Team / Mission:	______________________________ ____________		
Role / Position:	______________________________ ____________		

SpaceCDF Tab: Functions

Key Equations Reference

> *Function Types: Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support*
>
> *Universal Functions (every spacecraft):*
>
> *1. Generate electrical power*
>
> *2. Maintain thermal environment*
>
> *3. Survive launch environment*
>
> *4. Communicate with ground (TTC)*
>
> *5. Dispose at end of life*
>
> *Coverage Rule: Every leaf function must trace to at least one requirement. No gaps.*
>
> *Allocation Rule: Multi-allocated functions create interfaces that must be explicitly managed.*

Part A: Mission Function Tree (20 min)

Draw your mission's complete function tree. Use minimum 3 levels of decomposition. Include both mission-specific and universal functions.

Mission-specific functions:

```
F-001: _______________________________________________
  +-- F-002: _______________________________________________
```

```
        +-- F-002a: ___________________________________________
        +-- F-002b: ___________________________________________
    +-- F-003: _______________________________________________
        +-- F-003a: ___________________________________________
    +-- F-004: _______________________________________________
    +-- F-005: _______________________________________________
```

Universal functions:

```
  F-010: Generate electrical power
    +-- F-011: _______________________________________________
    +-- F-012: _______________________________________________
  F-020: Maintain thermal environment
    +-- F-021: _______________________________________________
  F-030: Survive launch environment
  F-040: Communicate with ground (TTC)
  F-050: Dispose at end of life
```

Part B: Function-to-Requirement Traceability (15 min)

For each leaf function, write one derived requirement with a measurable threshold:

Part C: Multi-Allocation Analysis (10 min)

Identify at least one function that is allocated to more than one subsystem:

Function: ______________________________

Subsystem A (primary): ______________________________

Subsystem B (contributor): ______

Interface created between A and B: ______

Who "owns" this function? ______

What interface requirements are needed? ______

Part D: Performance Criteria (10 min)

For 4 key functions, define quantitative performance criteria:

Part E: SpaceCDF Coverage Check (15 min)

After entering functions in SpaceCDF:

1. How many total functions in your tree? ____
2. How many leaf functions? ____
3. How many leaf functions have NO linked requirements (amber badge)? ____
4. List the uncovered functions and derive requirements for them:
5. Are there any functions the tool generated that are NOT appropriate for your mission? List and explain why you removed them:

Notes & Reflections

Worksheet 2.3: Orbit Selection and Mission Architecture

Function ID	Function Name	Allocated To (Subsystem)	Derived	Type
Function	Performance Criterion	Value	Unit	Source
Function ID	Function Name	Derived Requirement		
Name:	____			
Date:	____			
Team / Mission:	____			
Role / Position:	____			

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

Key Equations Reference

> *Orbital period: T = 2pi sqrta^3/u   where a = R_E + h, u = 3.986 x 10^14 m^3/s^2, R_E = 6371 km*
>
> *Orbital velocity: v = sqrtu/a*
>
> *Eclipse fraction: f_ecl = (1)/(pi) \arccos(\fracsqrta^2 - R_E^2a)*
>
> *Sun-synchronous inclination: cos(i) = -\frac2\dot\Omega a^7/23 R_E^2 J_2 sqrtu   where J_2 = 1.0826 x 10^-3*
>
> *Hohmann D V: D V_1 = sqrtu/r_1(sqrt2r_2/(r_1+r_2) - 1)*
>
> *Swath width: W = 2htan(theta)*

Part A: Orbital Mechanics Calculations (20 min)

Your selected orbit: Altitude h = ____

Show all working for each calculation:

1. Semi-major axis:

a = R_E + h = 6371 + ____________________

2. Orbital period:

T = 2pisqrta^3/u = 2pisqrt ____________________

3. Orbital velocity:

v = sqrtu/a = sqrt3.986 x 10^14 /\ ____________________

4. Eclipse fraction (maximum):

f = (1)/(pi)\arccos(\fracsqrta^2 - R_E^2a) = (1)/(pi)\arccos(\fracsqrt\ \ ) = ____________________

5. Sunlight and eclipse time per orbit:

t_sun = T x (1 - f) = ____________________

t_ecl = T x f = ____________________

6. Sun-synchronous inclination (if applicable):

cos(i) = ____________________

Does this match your selected inclination? Y / N

Part B: Orbit Trade Matrix (10 min)

From SpaceCDF's Orbit Trade Advisor, record the top 3 candidates:

Rank	Orbit Type	Alt (km)	Inc (deg)	GSD	Revisit (d)	Natural Lifetime (yr)	FCC 5-yr?	Score
1								
2								
3								

Selected orbit: ____________________

Part C: Debris Compliance (10 min)

For your selected orbit:

Parameter	Value
Natural orbital lifetime	_____ years
FCC 5-year compliant?	Y / N
IADC 25-year compliant?	Y / N
Propulsion needed for deorbit?	Y / N
If yes, estimated deorbit D V	_____ m/s
Deorbit method (if needed)	____________________________________

Show D V calculation if propulsion needed:

Part D: Downstream Cascade Analysis (10 min)

How does your orbit selection affect each subsystem? Fill in the impact:

Subsystem	Orbit Parameter That Matters	Impact on Your Design
Payload (GSD)	Altitude = _____ km	GSD = _____ m
EPS (eclipse)	Eclipse fraction = _____	Eclipse duration = _____ min
Comms (link)	Slant range at 10 deg elev = _____ km	FSPL increase = _____ dB
Thermal	Eclipse/sunlight cycling	Hot case / Cold case concern: ______
AOCS	Orbit rate = _____ deg/min	Nadir tracking rate needed
Propulsion	Lifetime = _____ yr	Propulsion needed? Y / N
Radiation	TID estimate = _____ krad/yr	Electronics class: ____________

Part E: Discussion (10 min)

What would happen to your mission if you moved the orbit 100 km higher?

What would happen if you moved it 100 km lower?

Notes & Reflections

Worksheet 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

Name:	______________________________
Date:	______________________________
Team / Mission:	______________________________
Role / Position:	______________________________

SpaceCDF Tabs: System Architecture, Interfaces, Dashboard, Budget Breakdown

---

Key Equations Reference

> *Mass margin: Margin_% = \fracM_alloc - M_MEVM_alloc x 100%*
>
> *   Green: > 20% | Amber: 10--20% | Red: < 10%*
>
> *MEV: M_MEV = M_CBE + equipment margin + system margin*
>
> *Orbit-average power: P_avg = Sum (P_mode x f_duty)*
>
> *SA power: P_SA = P_peak + \fracP_ecl x t_eclt_sun x eta_charge*
>
> *ECSS Phase A margins: Equipment 20% + System 20% = ~44% compound*

Part A: Mass Budget (15 min)

Complete from your SpaceCDF design (or estimate if pre-run):

Subsystem	CBE Mass (kg)	% of Dry	Equip. Margin (20%)	MEV (kg)
Payload				
EPS				
AOCS				
Comms (TTC)				
OBC				
Thermal				
Structure				
Harness				
Propulsion				
Dry Total (CBE)		100%		
System margin (20%)				
Dry MEV				
Propellant				
Wet Mass				
Launcher Allocation				
Mass Margin				
Mass Margin %			G / A / R	

Show margin calculation:

Part B: Power Budget by Mode (15 min)

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
Comms (TX)					
Thermal					
Total					
Duty cycle (%)					

Orbit-average power calculation (show working):

P_avg = ______________________________

____

SA power required:

P_recharge = \fracP_ecl x t_eclt_sun x eta_charge = (\ \ \ \ x \ \ \ \ )/(\ \ \ \ x 0.9) = ______________

P_SA,EOL = P_peak + P_recharge = ______________________________

P_SA,BOL = P_SA,EOL / (1 - 0.025)^n = ______________________________

____

____

Part C: Interface Identification (15 min)

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types (M/E/T/D/R/O)	Key Concern
1				
2				
3				
4				
5				

Write 2 formal interface requirements for your most critical pair:

Interface: ______________________________

IR-001: ______________________________

____

IR-002: ______________________________

____

Verification method for IR-001: ______________________________

Part D: Conflict Resolution (10 min)

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict description: ______________________________

Severity: Critical / Major / Minor

Resolution options considered:

#	Option	Pros	Cons
1			
2			
3			

Selected resolution: ______________________________

Rationale: ______________________________

Part E: Budget Health Check (10 min)

From the SpaceCDF Dashboard, record all KPIs:

Budget	Value	Margin	Status (G/A/R)
Mass	_____ kg wet vs _____ kg alloc	_____%	
Power (worst mode)	_____ W vs _____ W SA	_____%	
Link	_____ dB margin	>= 3 dB?	
Cost	_____ MEUR vs _____ ceiling	_____%	
Pointing	_____ deg vs _____ deg req	_____%	

Which budget is tightest? ______________________________

What single design change would improve it? ______________________________

Notes & Reflections

Worksheet 3.1: Power System and Thermal Control Design

Name:	______________________________
Date:	______________________________
Team / Mission:	______________________________
Role / Position:	______________________________

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

Quick Reference: Power and Thermal Concepts

What is a Solar Array?

A solar array converts sunlight into electrical power using photovoltaic cells. In space, the solar flux (called the solar constant) is S = 1361 W/m^2 at 1 AU from the Sun -- about 40% more intense than at Earth's surface because there is no atmosphere absorbing part of the spectrum.

Solar cell types and efficiency (AM0, space spectrum):

Cell Technology	Typical Efficiency	Degradation Rate	Cost Relative	Notes
Silicon (Si)	16--18%	~3.5%/yr	Low	Legacy; rarely used in new missions
Gallium Arsenide (GaAs) single-junction	22--24%	~2.5%/yr	Medium	Good radiation tolerance
Triple-junction GaAs (InGaP/GaAs/Ge)	28--30%	~2.5%/yr	High	CubeSat standard today
Multi-junction (4J, 5J)	32--35%	~2%/yr	Very high	Emerging; used on flagship missions

For CubeSat design, assume triple-junction GaAs cells at 29.5% efficiency (eta = 0.295). Cells are mounted on panels with a packing factor of f_pack = 0.80--0.85, meaning 80--85% of the panel area is actually covered by active cells (the rest is gaps, wiring, and structural margin).

Degradation: Solar cells lose efficiency over time due to radiation damage (proton and electron bombardment in the Van Allen belts). At a rate of delta = 2.5% per year, after n years the remaining efficiency is (1 - 0.025)^n. This means a 3-year mission loses about 7.3% of its beginning-of-life (BOL) power, and a 5-year mission loses about 11.9%.

Body-Mounted vs Deployable Solar Arrays

Configuration	Advantages	Disadvantages	Typical Power
Body-mounted	No moving parts, no deployment risk, no stowed volume penalty	Limited area (only the sunlit faces), power varies with attitude	1U: ~2 W, 3U: ~7 W, 6U: ~12 W
Single deployable	More area, moderate complexity	One mechanism, partially blocks FOV	1U: ~4 W, 3U: ~15 W, 6U: ~30 W
Dual deployable	Maximum area for the form factor	Two mechanisms, larger stowed volume, deployment risk	3U: ~25 W, 6U: ~48 W

Rule of thumb: If your total power demand (including battery recharge) exceeds what body-mounted panels can provide, you need deployable arrays. Compare your P_SA,BOL to the reference power values above.

How Batteries Work in Space

Batteries store electrical energy to power the spacecraft during eclipse (when the satellite is in Earth's shadow and receives no sunlight). In LEO, eclipses occur every orbit -- typically 30--36 minutes out of a ~95-minute orbit period.

Key battery parameters:

- Depth of Discharge (DOD): The fraction of total battery capacity used each cycle. Lower DOD dramatically extends cycle life but requires a larger (heavier) battery.
- Cycle life: The number of charge/discharge cycles before the battery degrades below 80% of its original capacity.
- Specific energy: Energy stored per unit mass (Wh/kg). For space-grade Li-ion 18650 cells: 150--200 Wh/kg.

DOD	Typical Cycle Life (Li-ion 18650)	Suitable Mission Duration
80%	~500 cycles	Short missions (< 1 month)
50%	~2,000 cycles	Medium missions (< 6 months)
30%	~10,000 cycles	Multi-year LEO missions
20%	~30,000 cycles	Long-life LEO (> 5 years)

Design guidance: For a multi-year LEO mission (15 orbits/day), use DOD = 30%. For a 6-month technology demonstration, DOD = 50% is acceptable.

What Does the EPS Do?

The Electrical Power System (EPS) is the spacecraft's power utility. A typical CubeSat EPS board (e.g., GomSpace P31u) includes:

- MPPT (Maximum Power Point Tracking): A DC-DC converter that continuously adjusts the operating voltage of the solar array to extract maximum power. Efficiency: 90--95%.
- Battery charge controller: Manages battery charging with voltage and current limits to protect battery life.
- Regulated voltage rails: Provides fixed voltages (typically 3.3 V, 5 V, and unregulated battery voltage) to all subsystems.
- Switched outputs: Allows the OBC to turn subsystems on/off for power management and safe mode.
- Overcurrent protection: Protects against short circuits in any subsystem.

Thermal Control Basics

In space, there is no air for convection. Heat transfer occurs only by radiation and conduction. A spacecraft reaches thermal equilibrium when absorbed heat (solar flux, Earth albedo, Earth IR, internal electronics waste heat) equals radiated heat (to deep space at ~3 K).

ECSS thermal margins (ECSS-E-ST-31C): Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC. If the predicted temperature of any component is within 5 degC of its operating limit, you must add thermal control (heater, coating change, or radiator).

Key Equations Reference

> *SA EOL power: P_SA,EOL = P_peak + \fracP_ecl x t_eclt_sun x eta_charge*
>
> *SA BOL power: P_SA,BOL = \fracP_SA,EOL(1 - delta)^n   (delta = 0.025/yr for GaAs)*
>
> *SA area: A = \fracP_BOLeta_cell x S x costheta x f_pack   (eta = 0.295, S = 1361 W/m^2, f_pack = 0.85)*
>
> *Battery capacity: C = \fracP_ecl x t_eclDOD x eta_discharge   (DOD = 0.30, eta = 0.95)*
>
> *Thermal equilibrium: T = (\fracQ_abs + Q_int\varepsilon sigma A_rad)^1/4   (sigma = 5.67 x 10^-8 W/m^2/K^4)*

> *ECSS thermal margins: Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC*

Worked Example: UniSat-1 (1U) Power and Thermal

Solar Array Sizing (Body-Mounted)

UniSat-1 uses body-mounted solar cells on 5 faces (the 6th face is the deployment interface). ISS orbit: 400 km altitude, 92.4 min period, 56 min sunlight, 36 min eclipse.

Given: $P_{peak,sunlight}$ = 1.2 W, $P_{eclipse}$ = 0.5 W (OBC only), mission lifetime = 6 months.

Step 1 -- Effective illuminated area (passive magnetic attitude, ~1.5 faces average):

$A_{eff} = 1.5 \times (0.10 \times 0.10) = 0.015$ m^2

Step 2 -- SA BOL power:

$P_{SA,BOL} = \eta \times S \times A_{eff} \times f_{pack} = 0.295 \times 1361 \times 0.015 \times 0.80 = 4.82$ W (illuminated peak)

Step 3 -- Orbit-average power available:

P_avg = P_SA,BOL x \fract_sunT_orbit x eta_EPS = 4.82 x (56)/(92.4) x 0.85 = 2.48 W

Step 4 -- After 6-month degradation:

$(1 - 0.025)^{0.5} = 0.987$, so $P_{avg,EOL} = 2.48 \times 0.987 = 2.45$ W

Step 5 -- Power demand: $P_{avg,demand}$ = 0.68 W. Margin = 2.45 - 0.68 = 1.77 W (72%). Comfortable.

Battery Sizing

Given: P_{ecl} = 0.5 W, t_{ecl} = 36 min = 0.60 h, DOD = 0.50 (acceptable for 6-month mission), eta = 0.95.

$C_{bat} = (0.5 \times 0.60)/(0.50 \times 0.95) = (0.30)/(0.475) = 0.63$ Wh

Specify minimum 10 Wh (standard GomSpace NanoPower P31u battery). Actual DOD per eclipse = 0.63/10 = 6.3%. Cycle count: 6 months at 15 orbits/day = 2,740 eclipses. At 6.3% actual DOD, battery life is not a concern. Pass.

Thermal Check (Passive Only)

UniSat-1 uses no heaters or MLI. At 400 km, strong Earth IR (~240 W/m^2) provides a warm floor. With COTS electronics rated -20 to +60 degC and a 400 km orbit, the thermal environment stays within approximately -10 to +45 degC for a 1U with aluminium/anodised surfaces. A simplified cold-case equilibrium calculation gives a very cold steady-state temperature, but the thermal mass of a 1 kg CubeSat limits actual cooling during 36-minute eclipses. No heaters required for a 6-month mission at 400 km.

Part A: Solar Array Sizing (20 min)

Show all calculations step by step.

1. Peak sunlight power demand: P_peak = ______________________

2. Eclipse power demand: P_ecl = ______________________

3. Eclipse and sunlight times (from Worksheet 2.3):

t_ecl = ______________________

4. Recharge power:

P_recharge = \fracP_ecl x t_eclt_sun x eta_charge = (\ \ \ \ \ x \ \ \ \ \ )/(\ \ \ \ \ x 0.9) = ______________

---

5. SA EOL required:

P_SA,EOL = P_peak + P_recharge = ______________________

6. EOL degradation factor (mission lifetime = ______________________

(1 - 0.025)^\ \ \ = ______________________

7. SA BOL required:

P_SA,BOL = \fracP_SA,EOL(1-delta)^n = (\ \ \ \ \ )/(\ \ \ \ \ ) = ______________

---

8. SA area:

A = \fracP_BOL0.295 x 1361 x cos(\ \ \ ) x 0.85 = (\ \ \ \ \ )/(\ \ \ \ \ ) = ______________

---

9. SA type needed (compare to CubeSat reference values):

- [ ] Body-mounted only (1U: ~2 W, 3U: ~7 W, 6U: ~12 W)
- [ ] Single deployable (1U: ~4 W, 3U: ~15 W, 6U: ~30 W)
- [ ] Dual deployable (3U: ~25 W, 6U: ~48 W)

10. SA mass:

m_SA = A x sigma = ______________________

(body-mounted: sigma = 2.5 kg/m^2; deployable: sigma = 1.5 kg/m^2)

Part B: Battery Sizing (10 min)

Show all calculations:

1. Eclipse energy:

E = P_ecl x \fract_ecl60 = ______________________

2. Battery capacity required:

C = (E)/(DOD x eta) = (\ \ \ \ \ )/(0.30 x 0.95) = (\ \ \ \ \ )/(0.285) = ______________________

3. With 20% margin: C_spec = ______________________

---

4. Battery mass:

m_bat = \fracC_specE_specific = (\ \ \ \ \ )/(150) = ______________________

5. Cycle life check:

Number of eclipses in mission = ______________________

At DOD = 30%, Li-ion provides ~10,000 cycles. Is this sufficient? Y / N

Part C: Thermal Analysis (15 min)

1. Identify hot case and cold case for your orbit:

Parameter	Hot Case	Cold Case
Solar illumination		
Operational mode		
Internal dissipation		
Coating condition	BOL / EOL	BOL / EOL
Key concern		

2. From SpaceCDF Dashboard, record thermal predictions:

Component	Min Predicted (degC)	Max Predicted (degC)	Operating Min (degC)	Operating Max (degC)
Payload / sensor				
Battery				
OBC				
Transponder				

3. Margin check (need >= 5 degC):

4. If any margin fails, what thermal control action would you take?

Part D: SpaceCDF Verification (15 min)

Compare your hand calculations to SpaceCDF values:

Component	Hot margin = Max_op - Max_pred	Cold margin = Min_pred - Min_op	Pass?
Parameter	Hand Calculation	SpaceCDF Value	Difference
SA power required (BOL)	_____ W	_____ W	_____ W
Battery capacity	_____ Wh	_____ Wh	_____ Wh
Orbit-average power	_____ W	_____ W	_____ W
Eclipse duration	_____ min	_____ min	_____ min

If there are significant differences, explain why:

Decision Justification

Explain WHY you chose each element of your power and thermal design. Consider the alternatives you rejected and the trade-offs involved.

Solar array configuration choice and rationale:

Battery sizing rationale (why this DOD? why this capacity margin?):

Thermal control approach (passive only? heaters? coatings?):

Notes & Reflections

Worksheet 3.2: Attitude and Orbit Control System (AOCS)

Name:	______________________________

Date:	____
Team / Mission:	____
Role / Position:	____

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

Quick Reference: AOCS Concepts

What is AOCS?

The Attitude and Orbit Control System performs two distinct jobs:

- Attitude Determination (AD): Figuring out which way the spacecraft is pointing, using sensors.
- Attitude Control (AC): Changing or holding the spacecraft's orientation, using actuators.

These are separate problems. You can know your orientation perfectly (great sensors) but still wobble if your actuators are weak. Or you can have powerful actuators but poor knowledge of where you are pointing.

Sensors: How Do We Know Where We Are Pointing?

Sensor	What It Measures	Accuracy	Mass	Power	Cost	When You Use It
Magnetometer	Direction of Earth's magnetic field	~2--5 deg	~5 g	~0.1 W	~1 kEUR	Coarse determination; always on for B-field knowledge
Coarse sun	Direction to the Sun (1--2 axes)	~2--5 deg	~5 g	~0.01 W	~0.5 kEUR	Safe mode; initial acquisition
Fine sun sensor	Direction to the Sun (2 axes, high precision)	~0.1--0.5 deg	~10 g	~0.05 W	~2 kEUR	Sun-pointing missions; backup to star tracker
Star tracker	Full 3-axis attitude from star patterns	~0.001--0.01 deg (3--30 arcsec)	50--350 g	~0.5--1.5 W	~20--35 kEUR	Precision pointing for imaging, comms, science
MEMS gyroscop	Angular rate (rotation speed)	Drift: ~1--10 deg/hr	~20 g	~0.3 W	~3 kEUR	Fills gaps during star tracker blinding; slew control
Earth sensor	Nadir direction (local vertical)	~0.1--0.5 deg	~50 g	~0.5 W	~5 kEUR	Nadir-pointing missions (less common on CubeSats)

Key insight: Star trackers are by far the most accurate sensor. They work by photographing the sky, matching the observed star pattern to an onboard catalogue, and computing the spacecraft's orientation to within a few arcseconds. However, they are blinded by the Sun, Moon, or Earth limb entering their field of view, so sun sensors or gyroscopes provide backup during these periods.

Actuators: How Do We Control Our Orientation?

Actuat	How It Works	Torque	Precision	Mass	Power	When You Use It
Permanent						

magnet

Aligns spacecraft with Earth's B-field like a compass needle	~10^-5 N m

(restori
ng)

~10--15 deg	~20 g	0 W	Passive stabilisation for missions with no pointing need

Hysteresis rods	Dissipate rotational energy through magnetic hysteresis; damp tumbling	N/A (damping	N/A	~10 g each	0 W	Always paired with permanent magnet for passive stabilisation
Magnetorquers (MTQ)	Electromagnetic coils that interact with Earth's magnetic field to produce torque	~10^-6--10^-4 N m	~2--5 deg (as primary); N/A (as	~10--30 g each	0.1--0.2 W	Primary control for low-precision missions; desaturation of reaction wheels for all RW-based systems
Reaction wheels	Spinning flywheels that exchange angular momentum with the spacecraft	~10^-4--10^-2 N m	~0.001--0.01 deg	30--120 g each	0.3--1.5 W	Precision pointing -- any mission requiring < 2 deg accuracy
Thrusters	Expel mass (cold gas, ions) to produce torque via offset force	Variable	~0.1--1 deg	System-dependent	High	Large slews; orbit adjust; when MTQs are insufficient (deep space)

Magnetorquers -- two critical roles:

1. As primary actuators (missions needing only 2--5 deg pointing): MTQs alone provide 3-axis control, but only in LEO where Earth's magnetic field is strong. They cannot produce torque parallel to the local field vector, so control authority varies around the orbit.
2. As desaturation actuators (all RW-based missions): Reaction wheels absorb disturbance torques by spinning up. Over time, they accumulate angular momentum and reach saturation (maximum spin speed). MTQs "dump" this stored momentum by generating opposing torque against Earth's magnetic field. This is done approximately once per orbit, taking 5--15 minutes.

Reaction wheels -- why 4 wheels? Three wheels (one per axis) provide minimum 3-axis control. A 4th wheel on a skew axis provides single-fault tolerance: if any one wheel fails, the remaining three still provide full 3-axis control.

AOCS Architecture Decision Table

Use this table to select your architecture based on your pointing requirement:

Pointing Requirement	Architecture	Sensors	Actuators	Mass	Power	Cost
> 5 deg (no pointing need)	Passive magnetic	Magnetometer	Permanent magnet + hysteresis rods	~50 g	0 W	~2 kEUR
2--5 deg	Magnetorquer-based	Coarse sun sensors + magnetometer	3-axis MTQ	~100 g	0.2 W	~8 kEUR
0.1--2 deg	Reaction wheel system	Fine sun sensors + magnetometer	3--4 RW + 3 MTQ (desaturation)	~500 g	2--4 W	~35 kEUR
< 0.1 deg	Fine pointing	Star tracker + fine sun sensors	4 RW + 3 MTQ	~800 g	3--5 W	~55 kEUR
< 0.01 deg	Very fine pointing	Star tracker + MEMS gyro + sun sensors	4 RW + 3 MTQ	~1200 g	4--6 W	~80 kEUR

How to use this table: Find your mission's pointing requirement (from your requirements worksheet). Read across to find the architecture, sensors, and actuators you need. If your mission is an Earth-observation camera needing 0.05 deg pointing, you need fine pointing (star tracker + 4 RW + 3 MTQ). If your mission is an IoT relay needing only 5 deg pointing, magnetorquers alone suffice.

Key Equations Reference

> *Gravity gradient torque: T_gg = (3u)/(2a^3)|I_z - I_x|sin(2theta)*
>
> *Aerodynamic torque: T_aero = (1)/(2)rho v^2 C_D A * d_cp-cm*
>
> *SRP torque: T_SRP = (S)/(c) A_s (1+q) * d_sp-cm*
>
> *Magnetic torque: T_mag = M x B*
>
> *Momentum storage: H = T_dist x t_desat/2*
>
> *Pointing error (RSS): theta_total = sqrtSum theta_i^2*
>
> *MTQ torque: T_MTQ = m_dipole x B x sin(alpha)*

Worked Example: UniSat-1 (1U) AOCS

Architecture Selection

UniSat-1 carries a MEMS magnetometer payload that does not require accurate pointing. In fact, a slowly rotating/tumbling state provides magnetic field measurements across multiple directions, improving data quality.

Architecture: Passive magnetic stabilisation (permanent magnet + hysteresis rods). No pointing error budget is needed because there is no pointing requirement.

Parameter	Value
Pointing accuracy	~10--15 deg (to local magnetic field)
Mass	~30--50 g
Power	0 W
Cost	~1--2 kEUR

Disturbance Torques at 400 km

Even though UniSat-1 has no active AOCS, understanding the disturbance environment confirms that the passive magnet provides sufficient restoring torque.

Gravity gradient (1U is nearly cubic, so $I_z \approx I_x$, making this very small):

$$T_{gg} \approx (3 \times 3.986 \times 10^{14})/(2 \times (6.771 \times 10^{6})^{3}) \times |I_z - I_x| \approx 1 \times 10^{-8} \text{ N m}$$

Aerodynamic (ρ at 400 km $\approx 4 \times 10^{-12}$ kg/m^3, $v \approx 7670$ m/s):

$$T_{aero} = 0.5 \times 4 \times 10^{-12} \times 7670^2 \times 2.2 \times 0.01 \times 0.01 \approx 3 \times 10^{-8} \text{ N m}$$

Permanent magnet restoring torque: \sim 1×10^{-5} N m -- three orders of magnitude larger than all disturbances. The satellite stays field-aligned.

Part A: AOCS Architecture Selection (10 min)

Pointing requirement from your mission: ______________________________

AOCS architecture selected (tick one):

- [ ] Passive magnetic (> 5 deg)
- [ ] Magnetorquers only (2--5 deg)
- [ ] Reaction wheels + MTQ (0.1--2 deg)
- [ ] Fine pointing: RW + star tracker + MTQ (< 0.1 deg)
- [ ] Very fine: RW + ST + gyro + MTQ (< 0.01 deg)

Justification for selection:

____

____

Estimated AOCS mass: ______________________________

Part B: Disturbance Torque Calculations (20 min)

Calculate at least 2 disturbance torques for your spacecraft and orbit. Show all working.

Spacecraft properties:

Parameter	Value	Unit
Mass		kg
Form factor		U
I_z (largest MOI)		kg m^2
I_x (smallest MOI)		kg m^2
Cross-section area (largest face)		m^2
d_cp-cm (CP-CM offset estimate)		m

Orbit altitude

km

1. Gravity gradient torque:

T_gg = (3 x 3.986 x 10^14)/(2 x (\ \ \ \ )^3) x |I_z - I_x| =

T_gg = __________

2. Aerodynamic torque (rho at __________

T_aero = (1)/(2) x __________

T_aero = __________

3. (Optional) Solar radiation pressure torque:

T_SRP = __________

4. (Optional) Residual magnetic dipole torque (M ~= __________

T_mag = __________

Summary:

Source	Torque (N m)
Gravity gradient	
Aerodynamic	
SRP	
Magnetic dipole	
Total (RSS or sum)	

Which disturbance dominates? __________

Part C: Reaction Wheel Sizing (10 min)

Skip this section if your mission uses passive magnetic or MTQ-only architecture.

1. Torque requirement (>= 2 x total disturbance):

T_RW,req = 2 x ______________________________

2. Momentum storage (desaturation interval = 1 orbit = ______________________________

H = T_dist x \fract_desat2 = ______________________________

3. Selected reaction wheel:

Parameter	Required	Selected Product
Torque	_____ mN m	_____ mN m
Momentum	_____ mN m s	_____ mN m s
Mass	--	_____ g
Power	--	_____ W
Quantity	4 (3+1 redundancy)	

Part D: Pointing Error Budget (15 min)

Skip this section if your mission uses passive magnetic stabilisation (no pointing requirement).

Complete the RSS pointing error budget:

Error Source	Value (deg)	Value^2
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter (RW)		
Orbit knowledge		
Timing error		
Sum of squares		
RSS Total = sqrtsum	= _____ deg	

Requirement: ______________________________

Margin: ______________________________

Does the pointing budget close? Y / N

Which error source dominates? ______________________________

What action would most improve the budget? ______________________________

Part E: SpaceCDF Verification (10 min)

Compare your hand calculations to SpaceCDF's Pointing Budget display:

Parameter	Hand Calculation	SpaceCDF	Match?
RSS pointing error	_____ deg	_____ deg	
Dominant contributor	____________	____________	
Margin to requirement	_____ %	_____ %	

If there are differences, explain:

Decision Justification

Explain WHY you chose your AOCS architecture. Consider the alternatives and what drove your decision.

Why this architecture and not a simpler one?

Why this architecture and not a more capable one?

What is the single biggest risk in your AOCS design?

Notes & Reflections

Worksheet 3.3: Communications and Link Budget Design

Name:	____
Date:	____
Team / Mission:	____
Role / Position:	____

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

Quick Reference: Communications Concepts

What is a Link Budget?

A link budget is an accounting statement for your communication link. It adds up every gain (transmitter power, antenna gain) and subtracts every loss (distance, atmosphere, pointing error) to determine whether the signal arriving at the receiver is strong enough to decode. Everything is computed in decibels (dB) so that multiplication and division become addition and subtraction.

Decibel refresher:

Conversion	Formula	Common Values
Watts to dBW	P_dBW = 10log_10(P_W)	2 W = +3.0 dBW; 1 W = 0 dBW; 0.5 W = -3.0 dBW
Kelvin to dBK	T_dBK = 10log_10(T_K)	150 K = 21.8 dBK; 600 K = 27.8 dBK
Ratio to dB	G_dB = 10log_10(G)	x2 = +3 dB; x10 = +10 dB; x100 = +20 dB

What Each Term in the Link Budget Means

Term	What It Is	Analogy
TX Power	Electrical power fed to the transmitter amplifier	How loud you shout
TX Antenna Gain (G_TX)	How well the antenna focuses the signal into a beam (vs radiating equally in all directions)	Using a megaphone vs shouting in all directions
TX Line Losses (L_TX)	Power lost in cables and filters between amplifier and antenna	Sound absorbed by the megaphone walls
EIRP	Effective Isotropic Radiated Power = P_TX + G_TX - L_TX. The total "signal strength leaving the spacecraft"	Total loudness heard from a distance
FSPL	Free Space Path Loss -- signal weakens as it spreads over a larger sphere. Depends on distance and frequency	Voice getting quieter the farther away you walk
Atmospheric Loss	Signal absorbed/scattered by Earth's atmosphere (rain, oxygen, water vapour)	Fog muffling sound
Pointing Loss	Signal loss because the antenna beam is not perfectly aimed	Megaphone pointed slightly off-target
Polarisation Loss	Loss when TX and RX antenna polarisations do not match	Holding the megaphone sideways when the listener expects vertical

RX Antenna Gain (G_RX)	Ground station antenna ability to collect signal. Larger dish = more gain	Listener using a large ear trumpet
System Noise Temp (T_sys)	Combined noise from antenna, sky, and receiver electronics. Lower = better	Background noise in the room
G/T	Receiver figure of merit = G_RX - T_sys (in dB). Captures how "good" the ground station is	Ear trumpet size minus room noise
C/N_0	Carrier-to-noise-density ratio -- the received signal strength relative to noise	Signal-to-noise ratio before decoding
E_b/N_0	Energy per bit to noise density -- the fundamental measure of link quality. Divides C/N_0 by data rate	How much energy is available per bit of information
Link Margin	How much E_b/N_0 you have above the minimum needed. Must be >= 3 dB	Safety factor for your communication

Frequency Band Comparison

Parameter	UHF (Amateur)	S-band	X-band	Ka-band
Frequency	435--438 MHz	2200--2290 MHz	8025--8400 MHz	25.5--27.0 GHz
Bandwidth available	20 kHz	5 MHz	375 MHz	1.5 GHz
Maximum data rate	< 20 kbps	0.1--10 Mbps	10--400 Mbps	100--2000 Mbps
FSPL at 500 km (nadir)	148 dB	163 dB	174 dB	184 dB
FSPL at 1300 km (10 deg elev)	158 dB	171 dB	182 dB	192 dB
Atmospheric loss	~0.1 dB	~0.5 dB	~1.0 dB	~3--10 dB (rain!)
Typical spacecraft antenna	Monopole (0 dBi)	Patch (6 dBi)	Horn/patch array (10--15 dBi)	Small dish/array (20+ dBi)
Ground antenna	Yagi (10--13 dBi)	3 m dish (35 dBi)	3 m dish (45 dBi)	1 m dish (45 dBi)
License type	IARU coordination (free)	ISED/FCC ($30--45K)	ISED/FCC + ITU (higher)	Complex ITU filing
License timeline	~6 months	6--12 months	12+ months	12--18 months
Equipment availability	Many COTS	Many COTS	Growing COTS	Emerging
Best for	TT&C, telemetry, simple missions	Medium data rate, standard EO	High data rate, EO/SAR	Very high data rate

How to choose: Start with your data rate requirement. If you need < 20 kbps and have non-commercial data, UHF amateur is free and simple. If you need up to 10 Mbps, S-band is the CubeSat workhorse. For high-resolution imaging producing GBs per day, you need X-band or Ka-band.

Antenna Types

Antenna Type	Gain	Beamwidth	Pointing Need	Mass	Typical Band	Use Case
Monopole/dipole	~0 dBi	Omnidirectional	None	5--20 g	UHF	TT&C, simple telemetry
Patch (single)	~5--8 dBi	~70--90 deg	Low (~10 deg)	10--50 g	S-band	Standard CubeSat downlink
Patch array (2x2)	~12 dBi	~30--40 deg	Medium (~5 deg)	30--100	S/X-band	Higher-rate downlink
Horn	~10--15 dBi	~20--30 deg	Medium (~5 deg)	50--200 g	X-band	High-rate downlink
Parabolic reflector	~20--35 dBi	~2--10 deg	Tight (< 1 deg)	200--500 g	X/Ka-band	Very high-rate, DTE
Phased array	~15--25 dBi	Electronically steered	Electronic (fast)	100--300 g	S/X/Ka-band	Agile beam, multiple targets

Trade-off: Higher gain antennas provide stronger signals (enabling higher data rates or lower TX power), but they have narrower beams, which means the spacecraft must point more accurately. An omnidirectional antenna works regardless of spacecraft orientation but provides very low gain.

Modulation and Coding

The modulation scheme and forward error correction (FEC) code determine the minimum E_b/N_0 required for a target bit error rate (BER):

Modulation + Coding	E_b/N_0 Required (BER 10^-6)	Spectral Efficiency	Best For
BPSK uncoded	10.5 dB	1.0 bps/Hz	Legacy telecommand
QPSK uncoded	10.5 dB	2.0 bps/Hz	Simple telemetry
QPSK + convolutional (r=1/2)	5.0 dB	1.0 bps/Hz	Standard CCSDS TM
QPSK + LDPC (r=1/2)	2.0 dB	1.0 bps/Hz	High-efficiency downlink
QPSK + LDPC (r=3/4)	4.0 dB	1.5 bps/Hz	Balanced (CubeSat standard)
8PSK + LDPC (r=3/4)	6.5 dB	2.25 bps/Hz	High-rate downlink

Design guidance: For most CubeSat missions, QPSK + LDPC (r=3/4) at E_b/N_0 = 4.0 dB is the standard choice, offering a good balance of coding gain and implementation simplicity.

Key Equations Reference

> *EIRP: EIRP = P_TX + G_TX - L_TX   (dBW)*
>
> *FSPL: FSPL = 20log_10((4pi d f)/(c))   (dB)*
>
> *G/T: G/T = G_RX - 10log_10(T_sys)   (dB/K)*
>
> *C/N_0: C/N_0 = EIRP - FSPL - L_losses + G/T + 228.6   (dBHz)*
>
> *E_b/N_0: E_b/N_0 = C/N_0 - 10log_10(R_b)   (dB)*
>
> *Link margin: Margin = E_b/N_0,avail - E_b/N_0,req - L_impl >= 3 dB*
>
> *Antenna gain: G = eta_a(pi D/lambda)^2*
>
> *dB conversion: P_dBW = 10log_10(P_W);   2 W = +3.0 dBW; 1 W = 0 dBW*

Worked Example: UniSat-1 (1U) UHF Link Budget

UniSat-1 uses UHF amateur band at 437 MHz with 9600 bps downlink to an amateur ground station.

Step-by-Step Link Budget

Scenario: 400 km orbit, 437 MHz, 9600 bps, 10 deg minimum elevation (slant range ~1150 km), ground station with 13 dBi cross-Yagi (RHCP).

Line	Parameter	Calculation	Value	Unit
1	TX Power	0.5 W = 10log_10(0.5)	-3.0	dBW
2	TX Antenna Gain	Monopole	0.0	dBi
3	TX Line Losses	Short cable run	-0.5	dB
4	EIRP	-3.0 + 0.0 - 0.5	-3.5	dBW
5	FSPL	21.98 + 20log_10(1.15 x 10^6) + 20log_10(437 x 10^6) - 169.54	-155.5	dB
6	Atmospheric Loss	Minimal at UHF	-0.3	dB
7	Pointing Loss	Omni antenna, minimal	-0.5	dB
8	Polarisation Loss	RHCP-to-RHCP (cross-Yagi)	-0.5	dB
9	RX Antenna Gain	13 dBi cross-Yagi	+13.0	dBi
10	System Noise Temp	600 K (amateur + LNA) = 10log_10(600)	27.8	dBK
11	G/T	13.0 - 27.8	-14.8	dB/K
12	Boltzmann Constant		+228.6	dBW/K/
13	C/N_0	-3.5 - 155.5 - 0.3 - 0.5 - 0.5 + (-14.8) + 228.6	+53.5	dBHz
14	Data Rate	9600 bps = 10log_10(9600)	39.8	dBbps
15	E_b/N_0 available	53.5 - 39.8	+13.7	dB
16	E_b/N_0 required	QPSK + conv. code r=1/2	5.0	dB
17	Implementation Loss		2.0	dB
18	LINK MARGIN	13.7 - 5.0 - 2.0	+6.7	dB

Result: Link closes with 6.7 dB margin (> 3 dB). Pass.

FSPL Calculation Detail

FSPL = 21.98 + 20log_10(1.15 x 10^6) + 20log_10(437 x 10^6) - 169.54

= 21.98 + 121.21 + 172.81 - 169.54 = 146.46 ... wait, let us recompute:

20log_10(1.15 x 10^6) = 20 x 6.0607 = 121.2 dB

20log_10(437 x 10^6) = 20 x 8.640 = 172.8 dB

FSPL = 21.98 + 121.2 + 172.8 - 169.54 = 146.4 dB

Note: The line-5 value of 155.5 dB uses the slant range in metres (1.15 x 10^6 m) and frequency in Hz (437 x 10^6 Hz) in the single-formula form: 20log_10(4pi x 1.15 x 10^6 x 437 x 10^6 / 3 x 10^8) = 155.5 dB. Both approaches give the same result when applied consistently.

Data Throughput

At 4800 bps useful throughput (9600 bps channel rate with r=1/2 FEC), a 7-minute pass delivers:

V_pass = 4800 x 420 x 0.85 = 1.71 Mbit = 214 kB per pass

With 4 passes/day: ~0.84 MB/day. The magnetometer generates ~1.13 MB/day -- marginal. May need a second ground station or data prioritisation.

Part A: Frequency Band Selection (10 min)

Mission data rate requirement: ______________________________

License type selected: Amateur / Experimental / Commercial

Band selected: ______________________________

Rationale for band selection:

____

____

Licensing/filing required: ______________________________

Estimated licensing cost: ______________________________

Part B: Complete Link Budget (25 min)

Fill in ALL rows. Show computation for EIRP, FSPL, G/T, C/N_0, and margin.

Line	Parameter	Formula / Source	Value	Unit
1	TX Power	_____ W =		dBW
2	TX Antenna Gain			dBi
3	TX Line Losses			dB
4	EIRP = Line 1 + 2 + 3		______	dBW
5	Free Space Path Loss	20log_10(4pi x ___ x ___ /c)		dB
6	Atmospheric Loss			dB
7	Pointing Loss			dB
8	Polarisation Loss			dB
9	RX Antenna Gain			dBi
10	System Noise Temp	_____ K =		dBK
11	G/T = Line 9 - 10		______	dB/K
12	Boltzmann Constant		+228.6	dBW/K/H
13	C/N_0 = 4 + 5 + 6 + 7 + 8 + 11 + 12		______	dBHz
14	Data Rate	_____ bps = 10log_10(___)		dBbps
15	E_b/N_0 available = 13 - 14		______	dB
16	E_b/N_0 required (mod+code: ________)			dB
17	Implementation Loss			dB
18	LINK MARGIN = 15 - 16 - 17		______	dB

Show FSPL calculation:

FSPL = 20log_10(4pi) + 20log_10(d) + 20log_10(f) - 20log_10(c)

= 21.98 + 20log_10(______________________________

____

____

Does the link close (>= 3 dB)? Y / N

If margin is excessive, maximum achievable data rate at 3 dB margin:

R_b,max = 10^(E_b/N_0,avail - E_b/N_0,req - L_impl - 3)/10 x R_b = ______________

____

____

Part C: Data Budget (10 min)

Daily data generation:

V_gen = R_payload x t_imaging x N_orbits x f_compression

= ______________________________

Daily downlink capacity:

V_DL = R_DL x t_contact x N_passes x eta_protocol

= ______________________________

Data budget closes? (V_DL >= V_gen): Y / N

If no, what is your mitigation plan?

____

____

____

Part D: SpaceCDF Comparison (10 min)

Open the Link Budget tab, enter your parameters, and compare:

Parameter	Hand Calculation	SpaceCDF	Difference
EIRP (dBW)			
FSPL (dB)			
C/N_0 (dBHz)			
Link Margin (dB)			

If there are differences, explain:

Part E: Modulation and Coding Selection (5 min)

Selected modulation:

Selected FEC code:

Required E_b/N_0 at BER 10^-6:

Spectral efficiency:

Rationale:

Decision Justification

Explain WHY you chose your communications architecture. Address each decision explicitly.

Why this frequency band and not another?

Why this antenna type? What pointing does it require from AOCS?

Why this modulation/coding scheme? What trade-off did you make between data rate and link robustness?

Notes & Reflections

Worksheet 3.4: Structure, Propulsion, and Equipment Selection

Name:	________________________________
Date:	________________________________
Team / Mission:	________________________________
Role / Position:	________________________________

SpaceCDF Tabs: Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

Quick Reference: Structure and Propulsion Concepts

CubeSat Design Specification (CDS Rev 14) Key Dimensions

The CDS defines standard CubeSat form factors. All CubeSats must comply with these dimensions to fit inside standard deployers (e.g., ISIPOD, NRCSD, Exolaunch). The unit "U" represents a 10 x 10 x 11.35 cm cube.

Form Factor	Dimensions (mm)	Max Mass (kg)	Internal Volume (cm^3)	Rails	Typical Use
1U	100 x 100 x 113.5	2.0	~1,000	4	Technology demo, simple sensors
1.5U	100 x 100 x 170.2	3.0	~1,500	4	Extended 1U with more volume
2U	100 x 100 x 227.0	4.0	~2,000	4	IoT, AIS, simple payloads
3U	100 x 100 x 340.5	6.0	~3,000	4	Standard EO, comms, science
6U	100 x 226.3 x 340.5	12.0	~6,000	4	High-performance EO, SAR, comms
12U	226.3 x 226.3 x 340.5	24.0	~12,000	8	Advanced payloads, propulsion
16U	226.3 x 226.3 x 454.0	32.0	~16,000	8	Microsatellite-class missions

Key CDS requirements (all form factors):

Requirement	Specification	Why It Matters
Rail material	Hard anodised Al 7075-T6 or 6061-T6	Prevents cold welding in vacuum; provides structural load path
Rail cross-section	8.5 x 8.5 mm minimum contact	Transmits launch loads through deployer rails
Deployment switches	Minimum 1 per accessible rail face	Ensures spacecraft is powered off until deployed
RBF (Remove Before Flight) pin	Required	Physically disconnects batteries during ground handling
No protrusions beyond rail envelope (stowed)	Required	Prevents jamming inside the deployer
CG offset from geometric centre	<= 2 cm	Prevents uncontrolled tumbling at deployment; deployer balance
Fundamental frequency	> 40 Hz (first mode)	Avoids resonance with launch vehicle modes

Structural Loads and Margins

During launch, the spacecraft experiences:

- Quasi-static acceleration: 6--9 g axial (along rocket axis), 2--4 g lateral (sideways). Think of your 5 kg CubeSat feeling like it weighs 45 kg.
- Random vibration: Broadband shaking at 20--2000 Hz. Critical for PCB solder joints and connectors.
- Shock: Brief, high-g pulses (500--2000 g) at separation events. Critical for brittle components.

Margin of Safety (MoS): The ratio of what the structure can withstand to what it must withstand, minus 1. Must be >= 0. If MoS is negative, the structure fails under the design load.

Material properties (Al 7075-T6): Yield strength sigma_y = 503 MPa, Ultimate strength sigma_u = 572 MPa. Factor of Safety: 1.25 (yield), 1.5 (ultimate).

Propulsion Decision Tree

Not every CubeSat needs propulsion. Use this decision tree:

```
START: Does my mission need propulsion?
  |
  +-- Orbit altitude < 500 km?
  |     YES --> Natural atmospheric drag deorbits within 5 years (FCC compliant)
  |             Do I need orbit maintenance to extend lifetime?
  |               NO  --> NO PROPULSION NEEDED
  |               YES --> Need drag compensation (5-15 m/s per year)
  |
  +-- Orbit altitude 500-700 km?
  |     Natural lifetime is 5-25+ years
  |     FCC 5-year rule: likely NON-COMPLIANT without active deorbit
  |     --> PROPULSION REQUIRED for deorbit (50-150 m/s)
  |
  +-- Orbit altitude > 700 km?
  |     Natural lifetime is decades to centuries
  |     IADC 25-year guideline: NON-COMPLIANT
  |     --> PROPULSION REQUIRED for deorbit (high Delta-V)
  |
  +-- Constellation phasing needed?
  |     YES --> Add 10-50 m/s for orbit plane/phase adjustment
  |
  +-- Collision avoidance needed?
        YES --> Add 1-5 m/s reserve per manoeuvre
```

Key rule: Below ~500 km, you almost certainly do not need propulsion. Above ~600 km, you almost certainly do. The 500--600 km range depends on your specific ballistic coefficient and mission lifetime.

Propulsion Technologies Comparison

Type	How It Works	I_sp (s)	Thrust	Propellant Mass (100 m/s, 5 kg S/C)	System Dry Mass	Total System Mass	Burn Time	Cost
Cold gas	Pressurised gas expelled through nozzle	40--80	10--100 mN	0.87 kg	0.3 kg	1.17 kg	Minutes	~15 kEUR
Resistojet	Gas heated before expulsion	80--150	10--50 mN	0.34 kg	0.3 kg	0.64 kg	Minutes	~25 kEUR
Electrospray	Liquid metal ionised and accelerated by electric field	500--1500	0.01--1 mN	0.04 kg	0.9 kg	0.94 kg	Months	~50 kEUR
Hall effect	Xenon ionised and accelerated by magnetic/electric fields	800--1500	1--10 mN	0.05 kg	1.5 kg	1.55 kg	Weeks	~80 kEUR
Green monoprop	Catalytic decomposition of non-toxic propellant	200--250	0.1--1 N	0.21 kg	1.0 kg	1.21 kg	Seconds	~60 kEUR

The fundamental trade-off:

- High I_sp (electric propulsion): Uses very little propellant but has very low thrust, so manoeuvres take weeks or months. The thruster hardware is also heavier and more expensive.
- Low I_sp (chemical/cold gas): Uses much more propellant but provides high thrust for quick

manoeuvres. Hardware is simpler, lighter, and cheaper.

Choose based on: How much D V do you need? How fast must the manoeuvre happen? How much mass/volume/cost can you afford?

Key Equations Reference

> *Structural Margin of Safety: MoS = \fracsigma_allowsigma_design x FoS - 1 >= 0*
>
> *   FoS: Yield 1.25, Ultimate 1.5 (metallic)*
>
> *Tsiolkovsky rocket equation: D V = I_sp * g_0 * ln(m_0/m_f)*
>
> *Propellant mass: m_prop = m_dry x (e^D V/(I_sp * g_0) - 1)*
>
> *   g_0 = 9.80665 m/s^2*
>
> *Data storage: S = V_daily x N_days x f_safety*
>
> *Fundamental frequency: f_1 > 40 Hz (deployer requirement)*

Worked Example: UniSat-1 (1U) Structure and Propulsion

CDS Compliance

Parameter	1U Specification	UniSat-1 Design	Compliant
Dimensions	100 x 100 x 113.5 mm	100 x 100 x 113.5 mm (ISIS 1U frame)	Y
Maximum mass	2.0 kg (CDS) / 1.33 kg (NanoRacks)	1.0 kg target	Y
Rail material	Hard anodised Al 7075-T6	ISIS standard frame	Y
Deployment switches	Min 1 per accessible face	2 (ISIS standard)	Y
RBF pin	Required	Included	Y
CG offset	<= 2 cm	< 1 cm (symmetric layout)	Y
Protrusions (stowed)	None beyond rail envelope	UHF antenna stowed along rail	Y

Propulsion Decision

Orbit altitude: 400 km. Natural lifetime: ~8--14 months (ballistic coefficient BC = m/(C_D A) = 1.0/(2.2 x 0.01) = 45.5 kg/m^2).

FCC 5-year rule satisfied without propulsion. No orbit maintenance needed (6-month mission). No constellation phasing. Decision: No propulsion.

Structural Margin (Axial Load)

Given: Mass = 1.0 kg, axial load = 9 g, 4 rails.

F_axial = (1.0 x 9 x 9.81)/(4) = (88.3)/(4) = 22.1 N per rail

sigma = (22.1)/(8.5 x 10^-3 x 8.5 x 10^-3) = (22.1)/(7.225 x 10^-5) = 0.31 MPa

MoS_y = (503)/(0.31 x 1.25) - 1 = (503)/(0.39) - 1 = 1289 \gg 0 Pass (by a very large margin)

Key insight: Quasi-static axial stress is never the critical case for CubeSats. The real structural risks are: (a) PCB solder joint fatigue from random vibration, (b) deployment mechanism reliability, (c) stiffness (fundamental frequency > 40 Hz).

Equipment List

#	Category	Component	Mass (g)	Power (W)	Cost (kEUR)
1	Structure	ISIS 1U frame	200	--	4.0
2	EPS + Battery	GomSpace P31us	200	0.3	12.0
3	Solar cells	Body-mounted GaAs (5 faces)	50	--	8.0
4	OBC	Custom MSP430/Cortex-M	30	0.3	3.0
5	Comms	NanoCom AX100 (UHF)	60	0.5	8.0
6	Antenna	UHF monopole (deployable)	20	--	2.0
7	Payload	MEMS magnetometer PCB	50	0.2	5.0
8	AOCS	Permanent magnet + hysteresis rods	30	0	1.0
9	Harness	Internal cables, connectors	50	--	1.0
	TOTAL		690	~1.3 (peak)	~44

Mass margin: 1330 - 690 = 640 g (48%). With 20% equipment margin + 20% system margin (MEV): 1330 - 994 = 336 g (25%). Green.

Part A: CDS Compliance Check (10 min)

CDS Requirement	Your Design Value	Specification	Compliant?
Form factor: _____ U	Dimensions: _____ mm	_____ mm	Y / N
Maximum mass	_____ kg (wet)	_____ kg	Y / N
Rail material		Hard anodised Al	Y / N
Deployment switches	_____ (quantity)	Min 2	Y / N
RBF pin provided	Y / N	Required	Y / N
No protrusions beyond rails (stowed)	Y / N	Required	Y / N
CG within 2 cm of geometric centre	Estimated: _____ cm offset	<= 2 cm	Y / N
Fundamental frequency	Estimated: _____ Hz	> 40 Hz	Y / N

Non-compliance items and corrective actions:

Part B: Structural Margin Calculation (10 min)

Axial launch load: ______________________________

Load per rail (4 rails, equal sharing):

F_axial = \fracm x a_axial x g_04 = (\ \ \ \ x \ \ \ \ x 9.81)/(4) = ______________________________

Stress in rail (cross-section 8.5 x 8.5 mm = 7.225 x 10^-5 m^2):

sigma = (F)/(A) = (\ \ \ \ )/(7.225 x 10^-5) = ______________________________

Margin of safety (yield, Al 7075-T6, sigma_y = 503 MPa):

MoS = (503)/(\ \ \ \ x 1.25) - 1 = ______________________________

Pass? Y / N

Part C: Propulsion Decision (15 min)

Does your mission need propulsion?

Question	Answer
Orbit altitude	_____ km
Natural orbital lifetime	_____ years
FCC 5-year compliant without propulsion?	Y / N
Orbit maintenance required?	Y / N
Constellation phasing required?	Y / N
Active deorbit required?	Y / N

Decision: Propulsion / No propulsion

If propulsion is selected, complete this section. Show all working.

Total D V required:

Manoeuvre	D V (m/s)
Orbit maintenance (_____ yr)	
Deorbit	
Collision avoidance margin	
Total	

Propulsion type selected: ______________________________

Propellant mass (Tsiolkovsky):

m_prop = m_dry x (e^D V/(I_sp x g_0) - 1)

= ______________________________

System dry mass (thruster + tank + feed): ______________________________

Total propulsion system mass: ______________________________

Fraction of total spacecraft mass: ______________________________

Part D: Equipment Selection Log (25 min)

Using SpaceCDF's Equipment Browser, select components for each category:

Category	Component Selected	Manufacturer	Mass (kg)	Power	Cost (kEUR)	Qty	Total Mass
EPS Board						1	
Battery						1	
Solar Panels							
OBC						1	
Reaction Wheel						x	
Magnetorquer						x	
Star Tracker							
Sun Sensor						x	
Transponder						1	
Antenna							
Payload						1	
Structure						1	
Harness						1	
Propulsion							
TOTAL			____		____		

Parametric estimate (from Session 2.4): ______________________________

Equipment-based total: ______________________________

Difference: ______________________________

Any incompatibilities detected? (RF band mismatch, voltage mismatch, interface conflict)

Part E: Component Trade Study (15 min)

Subsystem traded: ______________________________

Criterion	Weight	Option A: _________	Score	Option B: _________	Score	Option C: _________	Score
Mass	0.20						
Power	0.15						
Cost	0.20						
TRL	0.15						
Heritage	0.15						
Performanc	0.15						
Weighted	1.00		_____		_____		_____

Winner: ______________________________

Rationale (beyond the numbers):

Part F: Final Budget Health Check (10 min)

After equipment selection, record final budget status:

Budget	Value	Margin	Status
Mass (wet vs allocation)	_____ / _____ kg	_____ %	G / A / R
Power (worst mode vs SA)	_____ / _____ W	_____ %	G / A / R
Link margin	_____ dB	>= 3 dB?	G / A / R
Cost vs ceiling	_____ / _____ MEUR	_____ %	G / A / R
Pointing (RSS vs req)	_____ / _____ deg	_____ %	G / A / R
Data (DL vs gen)	_____ / _____ GB/day	_____ %	G / A / R

Any budget that does not close? ______________________________

Proposed fix: ______________________________

Decision Justification

Explain WHY you made each key structural and propulsion decision.

Form factor selection rationale (why this size and not larger/smaller?):

Propulsion decision rationale (if no propulsion: why not? if propulsion: why this type?):

Equipment selection rationale (which component trade-offs were hardest? what compromises did you make?):

Notes & Reflections

Worksheet 4.1: Equipment Selection & Bill of Materials

Name: ______________________________

Mission Name: ______________________________

Part A: Make/Buy/Reuse Decisions

For each subsystem, document the procurement decision and rationale:

Subsystem	Decision (Buy/Reuse/Make)	Rationale	Lead Time
Structure			
EPS (solar array)			
EPS (battery)			
EPS (distribution)			
OBC			
AOCS (sensors)			
AOCS (actuators)			
TTC (transceiver)			
TTC (antenna)			
Thermal			
Payload			
Propulsion (if applicable)			

Part B: Bill of Materials (BOM)

Complete the BOM from the SpaceCDF Equipment Browser selections:

Part C: Budget Tracking

After all selections, record the live budget values from the SpaceCDF Dashboard:

Item ID	Component	Manufacturer	Part Number	Qty	Mass (g)	Power (W)	Cost (kEUR)	TRL	ECCN
STR-001									
EPS-001									
EPS-002									
EPS-003									

OBC-001									
AOCS-00									
AOCS-00									
AOCS-00									
TTC-001									
TTC-002									
THM-001									
PLD-001									
HRN-001									
Budget	BOM Total	Parametric Estimate	Allocation	Margin	Margin				
Dry mass	_____ g	_____ g	_____ g	_____ g	_____				
Peak	_____ W	_____ W	_____ W	_____ W	_____				
Total	_____ kEUR	_____ kEUR	_____	_____	_____				
Volume	_____% of ___U		100%	_____%					

Does everything fit within allocation? Y / N

If not, what would you change? ____

____

____

Part D: Export Control Assessment

List any components that may require export licences:

Component	ECCN / USML Category	Controlled?	Licence Required?	Action Needed
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	

Are there any ITAR-controlled items? Y / N

If yes, what TAA (Technical Assistance Agreement) or DSP-5 licence is needed?

____

____

Part E: Interface Compatibility Check

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Data Protocol OK?	RF Band Match?	Connector OK?	Notes
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	

Y / N

Y / N

Y / N / NA

Y / N

	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	

Part F: Interface Incompatibility Resolution

Describe one interface incompatibility found during selection and how you resolved it:

Issue: ____________________

Components involved: ____________________

Root cause: ____________________

Resolution chosen: ____________________

Impact on budget: ____________________

Notes & Reflections

What was the most difficult equipment selection decision? Why?

Worksheet 4.2: Verification & Validation Planning

Name: ______________________________

Mission Name: ______________________________

Part A: IADT Method Assignment

For 10 key requirements from your SpaceCDF V&V Matrix, assign the verification method and justify:

#	Req ID	Requirement Text (abbreviated)	Method (I/A/D/T)	Phase (B/C/D)	Level (Unit/Sub/Sys)	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Summary count: Analysis: ______________________________

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis (early confidence) AND test (final proof):

Part C: Environmental Test Sequence

For your selected launch vehicle, specify the complete proto-flight test sequence:

Launch vehicle: ______________________________

Requirement ID	Requirement Text	Analysis (Phase B)	Test (Phase C/D)	Why Both Needed?

Step	Test	Specification	Duration	Pass Criteria
1	Initial functional test	Full performance baseline	_____ hr	All parameters nominal
2	Sine vibration	_____ g, _____ - _____ Hz	_____ min/axis x 3	No resonance shift > 5%
3	Random vibration	_____ gRMS overall	_____ min/axis x 3	No structural damage
4	Post-vibe functional test	Same as step 1	_____ hr	Compare to baseline
5	Thermal vacuum -- hot	+ _____ C (predicted max + 10 C)	_____ cycles	Functional at hot extreme
6	Thermal vacuum -- cold	- _____ C (predicted min - 10 C)	_____ cycles	Functional at cold extreme
7	Post-TVAC functional test	Same as step 1	_____ hr	Compare to baseline
8	EMC test	Required? Y / N	_____ day	No interference
9	Mass measurement	Calibrated scale (+/- 0.01 kg)	30 min	M <= _____ kg
10	Deployer fit check	_____ deployer	2 hr	Fits; slides freely

Total test campaign duration estimate: ____________________

Part D: Vibration Test Profile

Record the random vibration PSD profile from your launch vehicle PUG:

Frequency (Hz)	PSD Level (g^2/Hz)	Slope
20		Start
20 - _____		+ _____ dB/oct
_____ - _____		Flat
_____ - 2000		- _____ dB/oct
2000		End

Overall gRMS (calculated): ____________________

Show your calculation:

Part E: TVAC Cycle Profile

Sketch or describe one TVAC cycle:

Hot extreme: + ____________________

Ramp rate: ____________________

Functional test at each extreme? Y / N   Duration of functional test: ____________________

Calculate time for one cycle:

t_cycle = ______________________________

= ______________________________

Total TVAC campaign: ______________________________

Part F: Waiver Identification

Are there any requirements that cannot be verified by the standard method? Document potential waivers:

Notes & Reflections

Which environmental test do you consider most critical for your mission? Why?

Worksheet 4.3: Risk, Interfaces & FMECA

Name: ______________________________

Mission Name: ______________________________

Part A: Risk Register

Identify at least 5 technical risks for your mission design. Score each on the 5x5 matrix.

Scoring reminder: L x C = Score. Low (1-4), Medium (5-9), High (10-15), Critical (16-25).

Req ID	Standard Method	Proposed Alternative	Justification	Risk				
#	Risk Description	L (1-5)	C (1-5)	Score	Category (L/M/H/C)	Mitigation Strategy	Owner (Position)	Post-Mitigation Score
1								
2								
3								
4								
5								
6								
7								

How many risks are High or Critical (before mitigation)? ____________________

How many remain High or Critical after mitigation? ____________________

Part B: 5x5 Risk Matrix Plot

Plot your risks on the matrix below. Write the risk number in the appropriate cell.

```
              Consequence ->
              1       2       3       4       5
    L   5 |       |       |       |       |       |
    i   4 |       |       |       |       |       |
    k   3 |       |       |       |       |       |
    e   2 |       |       |       |       |       |
    l   1 |       |       |       |       |       |
```

Part C: N-Squared (N^2) Interface Matrix

Fill in the interface matrix for your mission's subsystems. Use abbreviations:

- P = Power (electrical)
- D = Data (I2C, SPI, UART, CAN, RS-422)
- R = RF (coaxial)
- M = Mechanical (mounting, alignment)
- T = Thermal (conductive/radiative)

	EPS	OBC	AOCS	TTC	Payload	Structure
EPS	---					
OBC		---				
AOCS			---			
TTC				---		
Payload					---	
Structure						---

Total number of interfaces identified: ______________________________

Interface conflicts detected: (describe at least 2)

1. Conflict: ______________________________

Resolution: ______________________________

2. Conflict: ______________________________

Resolution: ______________________________

Part D: FMECA Table

Complete for 4 critical components:

Part E: Single-Point Failure Analysis

List all single-point failures in your design:

Component	Function	Failure Mode	Cause	Local Effect	System Effect	Severity (1-5)	Detection Method	Compensating Provision
#	Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation (if not acceptable)		
1				Y/N	Y/N			
2				Y/N	Y/N			
3				Y/N	Y/N			
4				Y/N	Y/N			
5				Y/N	Y/N			

Total SPFs identified: ______________________________

Part F: Reliability Calculation

Calculate the series reliability for your mission's critical chain (components that must all work for mission success):

**R_series = R_1 x R_2 x ... x R_n = ______

**R_series = ______

If you added redundancy to one component (e.g., dual deployment mechanism), recalculate:

**R_redundant = 1 - (1 - R)^2 = 1 - (1 - ______

**New R_series (with redundancy) = ______

**Improvement: from ______

Part G: Risk Discussion

What is the highest-risk item in your design? What would it take (cost, mass, schedule) to bring the risk to an acceptable level?

Notes & Reflections

Which single-point failure concerns you the most? Would you accept it, or is mitigation essential?

Worksheet 4.4: Cost Estimation & Design Review

Name: ____________________

Mission Name: ____________________

Part A: Mission Cost Estimate (WBS)

Complete using both parametric estimates (from SpaceCDF Cost tab) and bottom-up estimates (from BOM vendor quotes):

Compone	Individual Reliability (R_i)	Mission Duration Used			
WBS #	Element	Parametric (kEUR)	Bottom-Up (kEUR)	Difference	Notes
1.0	Programme Management				
2.0	Systems Engineering				
3.0	Mission Assurance				
4.0	Payload				
5.0	Spacecraft Bus Hardware				
	5.1 Structure & mechanisms				
	5.2 EPS				
	5.3 AOCS				
	5.4 TTC				
	5.5 OBC & data handling				
	5.6 Thermal				
	5.7 Propulsion				
	5.8 Harness				
6.0	Integration & Test				
7.0	Software (FSW + GSW)				
8.0	Launch Services				
9.0	Ground Segment				
10.0	Operations (_____ years)				
	TOTAL	_____	_____		

Part B: Confidence Levels

P50 estimate (your best estimate): ____________________

P70 estimate (P50 x 1.2): ____________________

P80 estimate (P50 x 1.3): ____________________

Show your reasoning for the multiplier (is 1.3 appropriate, or should it be higher for your mission?):

Part C: Constellation Cost (if applicable)

Number of satellites: ______________________

**b = ln(learning_rate) / ln(2) = ln(______________________

Unit #	Unit Cost (kEUR) = C_1 x N^b	Cumulative (kEUR)
1		
2		
4		
8		
_____ (total)		

Show your calculation for the total constellation cost:

Hardware: ______________________

Spares (______________________

Launch: ______________________

Ground segment: ______________________

Operations (______________________

PM/SE/MA: ______________________

**Total constellation cost: ______________________

**P80 constellation cost (x 1.3): ______________________

Part D: Cost Drivers

Top 3 cost drivers in your mission (from SpaceCDF Cost Breakdown):

Rank	Cost Element	Cost (kEUR)	% of Total	Why is it Expensive?
1				
2				
3				

What would you change to reduce total cost by 20%?

Part E: Design Review Readiness

From the SpaceCDF Gate Review tab, assess your readiness for SRR/PDR:

#	Gate Criterion	Status (Pass/Fail/Manual)	Evidence Location	If Fail: Action Needed
1	Mission need justified			
2	Requirements complete & traceable			
3	All budgets close with margin			
4	Equipment selected (BOM complete)			
5	Interfaces defined, conflicts resolved			
6	V&V approach defined			
7	Risks identified with mitigations			
8	Cost estimate with WBS			
9	ConOps defined			
10	Sustainability compliance			

Must-pass criteria met: ______________________________

Overall readiness: READY / NOT READY

Action items to achieve readiness:

1. ______________________________
2. ______________________________
3. ______________________________

Part F: Presentation Preparation

Outline your 12-minute design review presentation:

Section	Duration	Key Message	Evidence/Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Notes & Reflections

What is the most uncertain element in your cost estimate? How would you reduce the uncertainty?

Worksheet 5.1: Ground Segment & Operations Architecture

Name: ______________________________

Mission Name: ______________________________

Part A: Ground Station Contact Analysis

Calculate contact time and data volume per pass for your mission.

Input Parameters

Parameter	Value	Source
Orbital altitude (h)	_____ km	SpaceCDF orbit selection
Minimum elevation angle (epsilon)	_____ deg	Ground station spec (typically 5-10)
Downlink data rate (R)	_____ Mbps	SpaceCDF link budget
Protocol efficiency (eta)	_____ (typically 0.85-0.95)	CCSDS overhead estimate
Ground station latitude	_____ deg	Selected station
Earth radius (R_E)	6371 km	Constant

Calculations

Maximum slant range:

rho = R_E x (sqrt((h/R_E + 1)^2 - cos^2(epsilon)) - sin(epsilon))

rho = 6371 x (sqrt((______________________________

rho = ______________________________

Average pass duration (approximate):

T_avg ~ ______________________________

Data volume per pass:

V_data = R x T x eta = ______________________________

Part B: Data Budget Closure

Parameter	Value	Unit
Daily data generation (payload)		MB/day
Daily data generation (housekeeping)		MB/day
Total daily generation		MB/day
Passes per day (GS 1: _____________)		passes
Data per pass (GS 1)		MB
Subtotal downlink (GS 1)		MB/day
Passes per day (GS 2: _____________)		passes
Data per pass (GS 2)		MB
Subtotal downlink (GS 2)		MB/day
Total daily downlink capacity		MB/day
Data budget margin		MB/day
Margin percentage		%

Does the data budget close? Y / N

If not, what changes would close it? (check all that apply)

- [ ] Add another ground station at: ________________
- [ ] Increase data rate to: _____ Mbps (requires: ________________)
- [ ] Reduce payload data generation to: _____ MB/day (impact: ________________)
- [ ] Add onboard compression (ratio: _____:1)
- [ ] Use SatNOGS network for additional passes
- [ ] Other: ____

Part C: Ground Station Network Design

Station #	Location	Latitude	Antenna	Band	Data Rate	Passes/Day	Role
1							Primary TTC
2							Payload DL
3							Backup/SatNOGS

Total ground segment cost estimate: ____

Cost Element	Annual Cost (kEUR)	Notes
Antenna rental / ownership		
MCS software licence		
Network connectivity		
Operations staff (_____ FTE)		
Total annual		
Mission lifetime (_____ yr)		

Part D: LEOP Timeline

Construct the LEOP timeline for the first 72 hours after separation:

Time (UTC)	Sim Time	Activity	Success Criterion	Status
T+0	Separation	Deployment switches release		[ ]
T+30 min	Timer expires	Antenna deployment commanded		[ ]
T+_____		Beacon acquisition	Carrier lock	[ ]
T+_____		First HK telemetry	Data decoded	[ ]
T+_____		First uplink command	ACK received	[ ]
T+_____		SA deployment (if separate)	Power generation confirmed	[ ]
T+_____		ADCS initialisation	Attitude determination active	[ ]
T+_____		ADCS calibration	Pointing < _____ deg	[ ]
T+_____		Full duplex validation	Uplink + downlink at ops rate	[ ]
T+_____		Health assessment complete	All subsystems nominal	[ ]

Part E: Mission Operations Timeline (Gantt Chart)

Sketch the operations timeline from launch to end of life:

```
Phase:       LEOP  Commission  Early Ops    Nominal Operations        EOL
Week:     1  2  3  4  5  6  7  8  9  ... 26 ... 52 ... 104 ... 156
          |  |  |  |  |  |  |  |  |       |      |       |       |
Staffing: ________________________________________________
          24/7     16/7       8/5        8/5 or automated    8/5

Key milestones:
  L+___: First light
  L+___: Commissioning complete
  L+___: Orbit maintenance manoeuvre (if applicable)
  L+___: End of nominal mission
  L+___: Extended mission (if applicable)
  L+___: Passivation and deorbit
```

Notes & Reflections

What is the most challenging aspect of your ground segment design? What would you do differently with more budget?

Worksheet 5.2: Mission Operations Concepts

Name: ______________________________

Mission Name: ______________________________

Part A: Operational Modes

Define all operational modes for your spacecraft. Use the SpaceCDF ConOps Editor to enter these.

Verify: Does the power budget close in every mode?

Mode	Description	Power (W)	Data Rate	Entry Condition	Exit Condition
Safe					
Detumble					
Standby					
Science					
Downlink					
Eclipse					
Manoeuvre					
Mode	Power Consumption (W)	Power Available (W)	Margin	Closes?	
Safe				Y / N	
Detumble				Y / N	
Standby				Y / N	
Science				Y / N	
Downlink				Y / N	
Eclipse (battery only)				Y / N	

Part B: FDIR Rules

Define at least 5 FDIR rules for your spacecraft:

#	Fault Description	Detection Method	Thresho	FDIR Level (0-4)	Autonomous	Recovery
1						
2						
3						
4						
5						
6						

Can Safe Mode sustain the spacecraft indefinitely (power + thermal)? Y / N

If not, what is the maximum Safe Mode duration? ____________________

Can every autonomous FDIR action be overridden from ground? Y / N

If not, which cannot? ____________________

Part C: Nominal Procedure

Write one nominal procedure for your mission (science observation or data downlink):

Procedure ID: ____________________

Preconditions:

Step	Action	Expected Result	If Not Achieved
1			
2			
3			
4			
5			
6			
7			
8			

Post-conditions:

Part D: Contingency Procedure

Write one contingency procedure (e.g., safe mode recovery, communication loss recovery):

Procedure ID: ____________________

Trigger condition:

Severity level: Green / Yellow / Orange / Red

Step	Action	Expected Result	If Not Achieved	Timeout
1				
2				
3				
4				
5				
6				

Escalation: If this procedure fails, escalate to: ______________________

Post-conditions:

Part E: Operations Staffing

Estimate the staffing requirements for each mission phase:

Phase	Duration	Shifts/Day	Staff/Shift	Total FTE	Annual Cost (kEUR)
LEOP	_____ days	3 (24/7)			
Commissioning	_____ weeks	2 (16/7)			
Early Operations	_____ months	1 (8/5)			
Nominal Operations	_____ years	1 (8/5) or auto			
End of Life	_____ weeks	1 (8/5)			
Total					

Part F: Anomaly Scenario

Hypothetical scenario: Your satellite enters Safe Mode during a weekend. The next ground contact is Monday morning (40 hours away). Walk through your response:

1. What is the spacecraft doing in Safe Mode?

2. Will it survive 40 hours? (Check power and thermal)

3. What telemetry will you review first on Monday?

4. What is your recovery plan?

Notes & Reflections

How much autonomy should your spacecraft have? Where is the line between onboard decision-making and ground control?

Worksheet 5.3: Mission Simulation Day

Name:

Mission Name:

Your Role:

Simulation Event Log

Record ALL events, decisions, and actions in real-time during the simulation:

#	Sim Time	Real Time	Event / Observation	Action Taken	Result	Logged By
1	T+0		Separation			
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

LEOP Checklist

Check off each LEOP milestone as it is achieved during the simulation:

#	Milestone	Time Achieved	Status	Notes
1	Separation confirmed		[ ]	
2	30-min timer expired		[ ]	
3	Antenna deployed		[ ]	
4	Beacon acquired		[ ]	
5	First HK telemetry decoded		[ ]	
6	First uplink command ACK'd		[ ]	
7	Solar array deployed (if sep.)		[ ]	
8	ADCS detumble complete		[ ]	
9	ADCS calibration complete		[ ]	
10	Full duplex validated		[ ]	
11	All subsystems nominal		[ ]	

Anomaly Response Form 1

Complete this form for each injected anomaly during the simulation.

Field	Entry
Anomaly ID	ANO-SIM-001
Sim Time	
Real Time	

Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used:

If not, what changes are needed?

Anomaly Response Form 2

Field	Entry
Anomaly ID	ANO-SIM-002
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used:

If not, what changes are needed? ______________________________

Anomaly Response Form 3

Field	Entry
Anomaly ID	ANO-SIM-003
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment:

Diagnosis:

Action Taken:

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				

Final Result:

Procedure used: ______________________________

Commissioning Status

After the commissioning phase of the simulation, record subsystem checkout status:

Subsystem	Checkout Complete?	Performance vs Spec	Issues Found
EPS	[ ]	Nominal / Degraded / Failed	
OBC	[ ]	Nominal / Degraded / Failed	
AOCS	[ ]	Nominal / Degraded / Failed	
TTC	[ ]	Nominal / Degraded / Failed	
Thermal	[ ]	Nominal / Degraded / Failed	
Payload	[ ]	Nominal / Degraded / Failed	
Propulsion	[ ]	Nominal / Degraded / Failed / NA	

Nominal Operations Summary

After the nominal operations phase, record:

Metric	Planned	Actual (Simulated)	Variance
Science observations completed			
Data downlinked (MB)			
Anomalies encountered			
Ground contacts made			
Safe mode entries			

Simulation Debrief

What went well?

What could be improved?

Design weaknesses revealed by simulation:

Procedure changes needed:

FDIR gaps identified:

Notes & Reflections

What was the most stressful moment during the simulation? What did it teach you about operations?

Worksheet 5.4: Final Review & Presentations

Name: ______________________________

Mission Name: ______________________________

Part A: Final Design Summary

Record the final state of your design from the SpaceCDF Dashboard:

Parameter	Value	Unit	Margin	Margin %	Status (G/A/R)
Dry mass		kg			
Wet mass (if propulsion)		kg			
SA power (EOL)		W			
Battery capacity		Wh			
Link margin (worst case)		dB			
Pointing accuracy		deg			
Data balance		MB/day			
Total cost (P50)		kEUR			
Total cost (P80)		kEUR			
Debris compliance score		/100			
Mission lifetime		years			

Do all budgets close? Y / N If not, which fail? ______________________________

Number of unresolved conflicts: ______________________________

Part B: Documentation Checklist

Tick each document generated from SpaceCDF Exports tab:

Document	Generated?	Complete?	Notes
MRD (Mission Requirements Document)	[ ]	[ ]	
Technical Specification	[ ]	[ ]	
Verification Plan	[ ]	[ ]	
ConOps Document	[ ]	[ ]	
SEMP (Systems Engineering Mgmt Plan)	[ ]	[ ]	
IRD (Interface Requirements Document)	[ ]	[ ]	
Risk Management Plan	[ ]	[ ]	
Test Plan	[ ]	[ ]	
Bill of Materials	[ ]	[ ]	
ITU Filing Template	[ ]	[ ]	
RSSSA Filing	[ ]	[ ]	

Export Control Assessment

[ ]

[ ]

End-of-Life Report	[ ]	[ ]	
COPUOS Registration	[ ]	[ ]	

Total documents generated: ____________________

Part C: Presentation Outline

Prepare your 12-minute final design review presentation:

Section	Time	Key Points to Cover	Evidence / Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Presenter(s): ____________________

Backup slides prepared on: ____________________

Part D: Peer Review Evaluation

Complete this section while reviewing another team's presentation.

Team being reviewed: ____________________

Reviewers: ____________________

#	Criterion	Score (0-2)	Comments / Questions
1	Mission need clearly justified		
2	Requirements complete and traceable		
3	All budgets close with margin		
4	Equipment selected with justified trades		
5	Interfaces defined, conflicts resolved		
6	V&V approach defined		
7	Risks identified with mitigations		
8	Cost estimate with WBS and P80		
9	ConOps and ground segment designed		
10	Sustainability compliance		
	TOTAL	/ 20	

Presentation quality:

Criterion	Score (1-5)	Comments
Clarity of communication		
Use of evidence (data, not assertions)		
Ability to answer questions		
Time management		

Board recommendation: GO / NO GO / GO WITH ACTIONS

Action items for the presenting team:

#	Action	Responsible	Deadline
1			
2			
3			

Strongest aspect of this team's design:

Biggest area for improvement:

Part E: Lessons Learned

1. What was the most challenging design decision your team faced?

2. What would you do differently if starting the course over?

3. What does the design need next (Phase B activities)?

4. What was the most valuable SpaceCDF feature?

5. What feature was missing that you needed?

6. What concept from this course will you apply in your future work?

Part F: Self-Assessment Rubric

Rate your own learning across the three weeks of the course. Be honest -- this is for your own reflection, not grading.

Competency	Week 1 Start (1-5)	Now (1-5)	Evidence of Growth
Define a space mission starting from the problem, not the solution			
Write verifiable, traceable requirements (SMART format)			
Conduct a structured trade study with weighted criteria			
Size subsystems using parametric models and engineering budgets			
Select COTS equipment and verify interface compatibility			
Construct a verification matrix with IADT methods			
Assess risk using a 5x5 matrix and define mitigations			
Estimate mission cost using parametric and bottom-up methods			
Design a ground segment and calculate data budget			
Define operational modes, FDIR rules, and procedures			
Respond to anomalies using a systematic process			
Present a design to a review board with evidence			

Scale: 1 = No familiarity, 2 = Basic awareness, 3 = Can apply with guidance, 4 = Can apply independently, 5 = Can teach others

Overall self-assessment:

At the start of this course, I would describe my systems engineering ability as:

Now, I would describe it as:

One thing I am most proud of from this course:

Notes & Reflections

Final thoughts on the course, the design process, and what comes next: